

# CHEMISTRY IS NOT A MYSTERY

## ZOOM A LEVEL SEMINAR

### (Guide to discussed Qns)

|                      | SUGGESTED ANSWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Points to note                                                                                                        |
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| <b>QUESTION ONE;</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                       |
| 1(a)<br>(i)          | First electron affinity is the energy given out when <b>an electron</b> is added to <b>a gaseous atom</b> to form a <b>uninegatively charged gaseous ion</b> ✓<br>or;<br>The energy given out when <b>one mole of electrons</b> combines with <b>one mole of gaseous atoms</b> to form <b>one mole of uninegatively charged gaseous ions</b> ✓                                                                                                                                                                                                                                                     | <b>Everything singular(word mole appears no where.</b><br><br><b>Use of the word one mole makes everything plural</b> |
| (ii)                 | The <b>heat evolved</b> when <b>one mole of ions</b> of a less electropositive element is displaced from its solution by a more electropositive element. ✓                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>This process is exothermic. One mole of ions but not ion</b>                                                       |
| (b)<br>(i)           | The first electron affinity for oxygen is <b>negative (exothermic)</b> ✓because <b>heat is given out as an electron is added to a neutral gaseous oxygen atom</b> ✓. The incoming electron experiences a <b>greater attraction by the nucleus than it is repelled</b> ✓. However, after formation of a uninegatively charged gaseous ion, <b>repulsion occurs when a second electron is added</b> ✓. Therefore <b>energy must be absorbed</b> ✓ to add the <b>second electron</b> in order to overcome <b>repulsion</b> ✓. This makes the <b>second electron affinity positive (endothermic)</b> ✓ | <b>ACCEPT ;</b><br><br><b>...as a mole of electrons is added... and mark accordingly</b>                              |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                              |
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| (ii)       | <p style="text-align: center;"><b>Using an energy level diagram;</b></p> <p style="text-align: center;"><b>Potential energy (kJmol<sup>-1</sup>)</b></p> <p style="text-align: center;"><b>Using an energy level diagram;</b></p> <p style="text-align: center;"><b>Cu<sup>2+</sup>(g) + O<sup>2-</sup>(g)</b></p> <p style="text-align: center;"><math>\Delta H_{2nd\ EA}^{\theta} = +770</math></p> <p style="text-align: center;"><b>Cu<sup>2+</sup>(g) + O(g)</b></p> <p style="text-align: center;"><math>\frac{1}{2} \Delta H_{BDE}^{\theta}</math></p> <p style="text-align: center;"><math>\Delta H_{1st\ EA}^{\theta} = -140.9</math> <b>Cu<sup>2+</sup>(g) + O<sup>-</sup>(g)</b></p> <p style="text-align: center;"><b>Cu<sup>2+</sup>(g) + <math>\frac{1}{2}</math> O<sub>2</sub>(g)</b></p> <p style="text-align: center;"><math>\Delta H_{2nd\ IE}^{\theta} = +1958</math></p> <p style="text-align: center;"><b>Cu<sup>+</sup>(g) + <math>\frac{1}{2}</math> O<sub>2</sub>(g)</b></p> <p style="text-align: center;"><math>\Delta H_{1st\ IE}^{\theta} = +750</math></p> <p style="text-align: center;"><b>Cu(g) + <math>\frac{1}{2}</math> O<sub>2</sub>(g)</b></p> <p style="text-align: center;"><math>\Delta H_{sub,Cu}^{\theta} = +339.3</math></p> <p style="text-align: center;"><b>Cu(s) + <math>\frac{1}{2}</math> O<sub>2</sub>(g)</b></p> <p style="text-align: center;"><b>0</b></p> <p style="text-align: center;"><math>\Delta H_{f,CuO}^{\theta} = -155.9</math></p> <p style="text-align: center;"><b>CuO(s)</b></p> <p style="text-align: center;"><math>-4081.5 = \Delta H_{lattice}</math></p> | <p><b>The cycle should have well balanced equations</b></p> <p><b>Any ions present should be gaseous and have correct charges</b></p> <p><b>The chemical symbols of the elements must be correct</b></p> <p><b>State symbols must be shown and must be correct</b></p> <p><b>Enthalpy values need to be shown along with the diagram</b></p> |
|            | $\Delta H_{f,CuO}^{\theta} = \Delta H_{sub,Cu}^{\theta} + \Delta H_{1st\ IE}^{\theta} + \Delta H_{2nd\ IE}^{\theta} + \frac{1}{2} \Delta H_{BDE}^{\theta} + \Delta H_{1st\ EA}^{\theta} + \Delta H_{2nd\ EA}^{\theta} + \Delta H_{lattice}$ $-155.9 = 339.3 + 750 + 1958 + \frac{1}{2} \Delta H_{BDE}^{\theta} + (-140.9) + 770 + (-4081.5) \checkmark$ $\Delta H_{BDE}^{\theta} = 2(-155.9 - 339.3 - 750 - 1958 + 140.9 - 770 + 4081.5) \checkmark$ $\Delta H_{BDE}^{\theta} = +498.4 \text{ kJmol}^{-1} \checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>ESCA</b></p> <p><b>E-Expression</b></p> <p><b>S-Substitution</b></p> <p><b>C-Computation</b></p> <p><b>A-Answer</b></p> <p><b>Reject 498.4, +498.4 or if no units</b></p>                                                                                                                                                              |
| (c)        | $\text{CuO(s)} + \text{H}_2\text{SO}_4\text{(aq)} \longrightarrow \text{CuSO}_4\text{(aq)} + \text{H}_2\text{O(l)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                              |
| (d)<br>(i) | <p>(Grey powder dissolves, ) the <b>blue solution</b> <math>\checkmark</math> fades in colour until it <b>turns colourless</b> <math>\checkmark</math> and a <b>brown solid deposited</b> <math>\checkmark</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p style="text-align: right;">2</p>                                                                                                                                                                                                                                                                                                          |

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| (ii)                 | <p>A <b>known volume</b> (<math>V \text{ cm}^3</math>) of <b>copper(II) sulphate solution</b> of a <b>known concentration</b> (<math>C \text{ moldm}^{-3}</math>) is put in a <b>plastic beaker</b>✓</p> <p>The <b>initial temperature</b>, <math>T_1^\circ\text{C}</math> of the solution is measured and recorded from the <b>thermometer</b>✓</p> <p>An <b>excess of zinc powder</b>✓ is added quickly and carefully to the solution while <b>stirring</b>✓</p> <p>The <b>final steady temperature</b>, <math>T_2^\circ\text{C}</math> is read and recorded from the thermometer.✓</p> <p style="text-align: center;"><b>Assumptions:</b></p> <p><b>Density of solution is equal to density of water</b> = <math>1.0 \text{ gcm}^{-3}</math>✓</p> <p><b>Specific heat capacity of solution is equal to specific heat capacity of water</b> = <math>4.2 \text{ Jg}^{-1}\text{K}^{-1}</math>✓</p> <p><b>No heat is lost to the surroundings</b>✓</p> <p style="text-align: center;"><b>Treatment of results:</b></p> <p style="padding-left: 40px;">Temperature rise = <math>\Delta T = (T_2 - T_1)^\circ\text{C}</math>✓</p> <p style="padding-left: 40px;">Mass of solution = (volume <math>\times</math> density) = <math>(V \times 1) = V \text{ g}</math></p> <p style="padding-left: 40px;">Heat change = mass of solution <math>\times</math> specific heat capacity <math>\times</math> temperature rise</p> <p style="padding-left: 80px;">Heat change = <math>(V \times 4.2 \times (T_2 - T_1))\text{J}</math>✓</p> <p style="padding-left: 40px;">Moles of copper(II) ions displaced = <math>\left(\frac{V \times C}{1000}\right)</math>✓</p> <p style="padding-left: 40px;"><math>\text{Cu}^{2+}(\text{aq}) + \text{Zn}(\text{s}) \longrightarrow \text{Cu}(\text{s}) + \text{Zn}^{2+}(\text{aq})</math>✓</p> <p style="padding-left: 40px;"><math>\left(\frac{V \times C}{1000}\right)</math> moles of <math>\text{Cu}^{2+}</math> are displaced with evolution of <math>(V \times 4.2 \times \Delta T)\text{J}</math>✓</p> <p style="padding-left: 40px;">1 mole of <math>\text{Cu}^{2+}</math> is displaced with evolution of <math>\left(\frac{1000 \times (V \times 4.2 \times \Delta T)}{V \times C}\right)\text{J}</math>✓</p> <p style="padding-left: 40px;">Enthalpy of displacement of <math>\text{Cu}^{2+}</math> = <math>-\left(\frac{1000 \times (V \times 4.2 \times \Delta T)}{V \times C}\right)\text{Jmol}^{-1}</math>✓</p> <p style="padding-left: 40px;"><b>or Enthalpy of displacement of <math>\text{Cu}^{2+}</math></b> = <math>-\left(\frac{(V \times 4.2 \times \Delta T)}{V \times C}\right)\text{kJmol}^{-1}</math></p> | <p><b>Volume and concentrations should be specified</b></p> <p><b>Zinc should be in excess</b></p> <p><b>Units of temperature shd be specified</b></p> <p><b>Deny 1.5 marks if assumptions not given</b></p> <p><b>Accept molecular equation</b></p> <p><b>Reject if (-) is missing</b></p> |
| <b>QUESTION TWO;</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                             |

NEXT SEMINAR IS ON ;  
Sat 09/10/2021  
2:00-4:00 pm

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| 2   | <p><b>N.B In all your answers; names of any intermediates shd be stated, conditions for the reactions well specified and no formulae or equations shd be used. ANY WRONG REAGENT(S), WRONG OR MISSING CONDITION OR STATEMENT, MAKES EVERYTHING AFTER IT WRONG.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Accept<br/>-chlorine in tetrachloromethane (to form 1,2-dichloroethane)<br/>-heated for boiled<br/>-accept ethanolic KOH<br/>-reject ammonia solution or just ammonia<br/>-reject bromopropane</p>                                               |
| (a) | <p><b>Any temperature within the ranges given for reactions that are temperature specific is acceptable.</b></p> <p><b>You should avoid giving industrial methods of preparation if question is specific to laboratory methods.</b></p> <p><b>You are advised to always make a sidework plan of your synthesis using equations and then write your final answer on the sheets of paper in words. Using the equations when not required makes you lose all the marks.</b></p> <p><b>Pent-1-yne from ethene</b></p> <p>Ethene gas is bubbled through a <b>solution of bromine in tetrachloromethane</b>✓ to form <b>1,2-dibromoethane</b>✓, which is then <b>boiled</b>✓ with an <b>excess alcoholic solution of potassium hydroxide</b> ✓ to form <b>ethyne</b>✓. Ethyne is then reacted with <b>sodium in liquid ammonia</b>✓ to form <b>sodium acetylide</b>(ethenylsodium) ✓ which is then reacted with <b>1-bromopropane</b> ✓ to form pent-1-yne.</p> |                                                                                                                                                                                                                                                     |
| (b) | <p><b>Propanoic acid to propan-2-ol</b></p> <p>Propanoic acid is reacted with an <b>ethereal solution of Lithium tetrahydridoaluminate(III)</b> ✓ to form <b>propan-1-ol</b>✓. Propan-1-ol is heated with an <b>excess of concentrated sulphuric acid</b> ✓ at <b>180°C</b>✓ to form <b>propene</b>✓. Propene is then bubbled through <b>cold (0-15°C) concentrated sulphuric acid</b>✓ to form <b>isopropyl hydrogensulphate/ propan-2-yl hydrogensulphate</b>✓ which is then <b>heated/ warmed with water</b> ✓ to form propan-2-ol.</p> <p><b>N.B Propene can also be reacted with dry hydrogen bromide to form 2-bromopropane which is then heated with sodium hydroxide solution to form propan-2-ol</b></p>                                                                                                                                                                                                                                         | <p>Accept <b>diborane(B<sub>2</sub>H<sub>6</sub>) instead of LiAlH<sub>4</sub>.</b><br/><b>Acc any other Conc H<sub>3</sub>PO<sub>4</sub> or Al<sub>2</sub>O<sub>3</sub> and corresponding temps.</b><br/><b>Reject propanol and prop-2-ene</b></p> |
| (c) | <p><b>Benzene to phenol</b></p> <p>A mixture of benzene and <b>fuming sulphuric acid</b>✓ is <b>warmed</b> ✓ to 35-50°C to form <b>benzenesulphonic acid</b> ✓ which is <b>fused</b> ✓ with solid <b>sodium hydroxide</b> ✓ at <b>300-350°C</b>✓ to form <b>sodium phenoxide</b> ✓ that is then treated with <b>dilute hydrochloric acid</b> ✓ to form phenol.</p> <p><b>Or;</b></p> <p>Benzene is reacted with <b>chlorine</b>✓ in presence of <b>aluminium chloride</b> ✓ at room temperature to form <b>chlorobenzene</b>✓. Chlorobenzene is hydrolysed using <b>steam</b></p>                                                                                                                                                                                                                                                                                                                                                                         | <p><b>Acc use of conc. H<sub>2</sub>SO<sub>4</sub> for several hours (about 8 hrs) or reflux</b><br/><b>Acc dilute H<sub>2</sub>SO<sub>4</sub> for dil.HCl</b></p> <p><b>Reject bromination</b> 4<br/><b>Reject water instead of</b></p>            |

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| <p>✓and <b>silicon catalyst</b> ✓at <b>425°C</b> ✓to form phenol.</p> <p><b>Or;</b></p> <p>Benzene is reacted with <b>chlorine</b>✓ in presence of <b>aluminium chloride</b> ✓at room temperature to form <b>chlorobenzene</b>✓. Chlorobenzene is then hydrolysed using <b>aqueous sodium hydroxide</b> ✓at <b>360°C</b>✓ and a <b>pressure of 150 atm</b>✓ to form <b>sodium phenoxide</b> ✓that is then treated with <b>dilute hydrochloric acid</b> ✓to form phenol.</p> <p><b>Or;</b></p> <p>Benzene is reacted with <b>chlorine</b>✓ in presence of <b>aluminium chloride</b> ✓at room temperature to form <b>chlorobenzene</b>✓. Chlorobenzene is then reacted with <b>ammonia gas in presence of copper(I) oxide catalyst at 250°C and high pressure</b> ✓to form <b>aminobenzene/phenylamine</b> .✓ The phenylamine is reacted with <b>sodium nitrite in dilute hydrochloric acid at a temperatures less than 10°C</b>✓ to form <b>benzenediazonium chloride</b>✓, which is finally <b>warmed in dilute sulphuric acid( or dissolved in water and the aqueous solution warmed)</b> ✓to form phenol.</p> <p><b>Or;</b></p> <p>A mixture containing <b>equal molar quantities of concentrated sulphuric acid and nitric acid</b>✓ is added to benzene and the resultant mixture is then refluxed with in a water bath maintained at <b>55-60°C</b> ✓to form <b>nitrobenzene</b>✓. The nitrobenzene is then <b>heated in presence iron turnings and dilute (30%) hydrochloric acid</b>✓and the mixture <b>neutralized with sodium carbonate</b> ✓to form <b>phenylamine/aminobenzene</b>✓(or <b>nitrobenzene is treated with tin and concentrated hydrochloric acid followed concentrated hydrochloric acid at 60°C to form phenylamine</b>). The phenylamine is reacted with <b>sodium nitrite in dilute hydrochloric acid at a temperatures less than 10°C</b>✓ to form <b>benzenediazonium chloride</b>✓, which is finally <b>warmed in dilute sulphuric acid( or dissolved in water and the aqueous solution warmed)</b> ✓to form phenol.</p> <p><b>Or;</b></p> <p>Benzene and <b>propene</b>✓ are compressed to <b>30 atm</b> ✓and passed over <b>phosphoric(V) acid</b>✓ at <b>250°C</b> ✓to form <b>cumene</b> (1-methylethylbenzene) ✓.</p> | <p><b>steam</b></p> <p><b>Reject bromination</b><br/><b>Emphasise physical state of NaOH(aq)</b><br/><b>Acc dilute H<sub>2</sub>SO<sub>4</sub> for dil.HCl</b></p> <p><b>Reject bromination</b></p> <p><b>This is called the CUMENE process</b><br/><b>Accept...to</b></p> |
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|     | Cumene is then oxidized by <b>air(oxygen)</b> ✓ to form <b>cumene hydroperoxide</b> (1-methylethylbenzene hydroperoxide) ✓ which is then treated with <b>dilute sulphuric acid</b> ✓ to form phenol.                                                                                                                                                                                                       | <b>form its hydroperoxide</b><br><b>Reject</b><br><b>conc. H<sub>2</sub>SO<sub>4</sub> or just H<sub>2</sub>SO<sub>4</sub></b>                                       |
| (d) | <b>Phenylmethanal (benzaldehyde/ benzenecarbaldehyde) from benzene</b><br>Benzene is reacted with <b>iodomethane(or bromomethane or chloromethane)</b> ✓ in presence of anhydrous <b>aluminium chloride catalyst(or aluminium bromide or iron(III) bromide)</b> ✓ to form <b>methylbenzene</b> ✓. Methylbenzene is oxidized by <b>chromyl chloride(chromium(VI) dichloride dioxide)</b> ✓ to benzaldehyde. | <b>Accept use of MnO<sub>2</sub> or CrO<sub>3</sub> in presence of dilute H<sub>2</sub>SO<sub>4</sub> and heat.</b><br><b>Reject procedure through benzoic acid.</b> |
| (e) | <b>Benzenesulphonic acid from benzoic acid</b><br>Benzoic acid is fused with <b>soda-lime( calcium oxide slaked with sodium hydroxide)</b> ✓ and the mixture <b>heated</b> ✓ strongly to form <b>benzene</b> ✓. Benzene is then mixed with <b>fuming sulphuric acid</b> ✓ is <b>warmed</b> ✓ to 35-50°C to form benzenesulphonic acid.                                                                     | <b>Accept formation sodium benzoate and fusing it with sodalime</b>                                                                                                  |

### QUESTION THREE;

|             |                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                           |
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| 3(a)<br>(i) |                                                                                                                                                                                                                                                                                                            | <b>Axes should be labeled and with correct units and scales to award</b><br><br><b>X-axis must begin from 11 to award</b><br><br><b>-½ for each wrongly plotted point</b> |
| (ii)        | Melting point <b>increases from sodium to magnesium</b> ✓ because sodium have <b>giant metallic structures</b> ✓ with metallic bonds <b>whose strength increases with increase in number of electrons contributed per atom to the metallic bond</b> ✓. (Magnesium which contributes two electrons per atom | <b>Reject across the period</b><br><br><b>Axes should be</b>                                                                                                              |

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2:00-4:00 pm

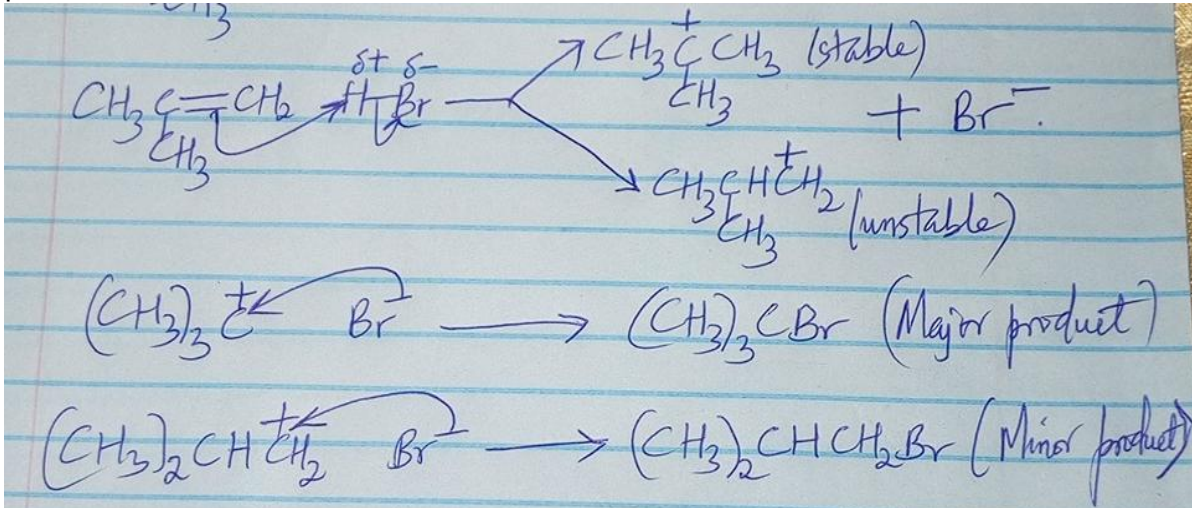
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|     | <p>forms stronger metallic bonds than sodium which contributes one electron per atom).</p> <p><b>Melting point increases further from magnesium to silicon because silicon has a giant molecular structure ✓ with very many strong covalent bonds ✓ that require a high amount of energy to break.</b></p> <p>The <b>decrease in melting point from silicon to phosphorus ✓</b> is because phosphorus has a <b>simple molecular structure ✓ with weak van der Waals forces of attraction ✓</b> that require a lower amount of energy to break.</p> <p>The <b>slight increase in melting point from phosphorus to sulphur ✓</b>, which also has a simple molecular structure is due to <b>increase in strength of the van der Waals forces with increase in molecular mass from phosphorus which is tetratomic to sulphur which is diatomic. ✓</b></p>                                   | <p><b>labeled and with correct units and scales to award Accept; silicon has highest mpt</b></p>                                                                          |
| (b) | <p><math>Si(s) + 2\overline{OH}(aq) + H_2O(l) \longrightarrow SiO_3^{2-}(aq) + 2H_2(g) \checkmark</math><br/> or <math>Si(s) + 2NaOH(aq) + H_2O(l) \longrightarrow Na_2SiO_3(aq) + 2H_2(g)</math></p> <p><math>P_4(s) + 3\overline{OH}(aq) + 3H_2O(l) \longrightarrow 3H_2PO_2^-(aq) + PH_3(g) \checkmark</math><br/> or <math>P_4(s) + 3NaOH(aq) + 3H_2O(l) \longrightarrow 3NaH_2PO_2(aq) + PH_3(g)</math></p> <p><math>3S(s) + 6\overline{OH}(aq) \longrightarrow 2S^{2-}(aq) + SO_3^{2-}(aq) + 3H_2O(l) \checkmark</math><br/> or <math>3S(s) + 6NaOH(aq) \longrightarrow 2Na_2S(aq) + Na_2SO_3(aq) + 3H_2O(l)</math></p> <p><math>S(s) + 6\overline{OH}(aq) \longrightarrow 2S^{2-}(aq) + S_2O_3^{2-}(aq) + 3H_2O(l)</math><br/> or <math>S(s) + 6NaOH(aq) \longrightarrow 2Na_2S(aq) + Na_2S_2O_3(aq) + 3H_2O(l)</math></p>                                                       | <p><b>Accept both ionic and molecular equations</b></p> <p><b>Deny ½ mark for wrong or missing state symbols</b></p> <p><b>Award no mark for unbalanced equations</b></p> |
| (c) | $NaH, \checkmark MgH_2, \checkmark SiH_4, \checkmark PH_3, \checkmark H_2S \checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Reject <math>PH_3, SH_2</math></b>                                                                                                                                     |
| (d) | <p>Sodium hydride reacts with water to form sodium hydroxide and hydrogen gas ✓<br/> <math>NaH(s) + H_2O(l) \longrightarrow NaOH(aq) + H_2(g) \checkmark</math></p> <p>Magnesium hydride reacts with water to form magnesium hydroxide and hydrogen gas ✓.<br/> <math>MgH_2(s) + 2H_2O(l) \longrightarrow Mg(OH)_2(aq) + 2H_2(g) \checkmark</math></p> <p>Silane reacts with water <b>in alkaline medium</b> to form silicate ions and hydrogen gas. ✓<br/> <math>SiH_4(g) + 2\overline{OH}(aq) + H_2O(l) \longrightarrow SiO_3^{2-}(aq) + 4H_2(g) \checkmark</math></p> <p><b>Phosphine does not react with water. ✓</b></p> <p>Hydrogen sulphide reacts with water to form a weakly acidic solution ✓<br/> <math>H_2S(g) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HS^-(aq) \checkmark</math><br/> or <math>H_2S(g) + 2H_2O(l) \rightleftharpoons 2H_3O^+(aq) + S^{2-}(aq)</math></p> | <p>7</p>                                                                             |

## QUESTION FOUR

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4<br>(a) | <p>In 2-Bromobutane, the bromine atom is <b>more electronegative</b>✓ than carbon atom hence attracts the bonding electrons more toward itself, gaining a <b>partial positive charge</b> whereas the <b>carbon atom gains a partial positive charge</b>✓. This makes the <b>carbon-bromine bond polar and weak hence easily broken</b>✓ to release <b>bromide ions in solution</b> and a <b>secondary carbocation</b>. The partial positive carbon being electron deficient, is <b>easily attacked by the nucleophilic hydroxide ion</b>✓ from sodium hydroxide to form <b>butan-2-ol</b>✓.</p> <p>However, in bromobenzene, the <b>lone pairs of electrons on the bromine atom interact with the delocalized pi electrons in the benzene ring</b>✓ making the <b>carbon-bromine bond very strong</b> ✓ and difficult to break.</p> | <p><b>The whole concept here is to explain why haloalkanes easily undergo nucleophilic substitution whereas arylhalides do not easily undergo such reactions</b></p>                  |
| (b)      | <p>2-Methylpropene combines with hydrogen bromide to form <b>bromide ions, a tertiary carbocation</b>✓ and a <b>primary carbocation</b>✓. The tertiary carbocation is <b>more stable</b> ✓ than the primary carbocation.</p> <p>Therefore the <b>amount of the tertiary carbocation is higher</b>✓ than that of the primary carbocation. The <b>most of the bromide ions</b>✓ combine with the tertiary carbocation to form 2-Bromo-2-methylpropane while a <b>few bromide ions</b>✓ combine with the primary carbocation to form 1-Bromo-1-phenylpropane as minor product.</p>                                                                                                                                                                 | <p><b>This is an explanation that requires you to explore stability of different of different types of carbocations. It also requires you to remember the Markownikoff's rule</b></p> |
| (c)      | <p style="text-align: center;"><b><u>Mg – 1s<sup>2</sup> 2s<sup>2</sup> 2p<sup>6</sup> 3s<sup>2</sup></u></b>✓<br/><b><u>Al – 1s<sup>2</sup> 2s<sup>2</sup> 2p<sup>6</sup> 3s<sup>2</sup> 3p<sup>1</sup></u></b>✓</p> <p>For magnesium, an electron is removed from a <b>completely filled 3s-sub energy level which is thermodynamically stable</b>✓. Higher amount of energy is <b>required to remove the electron</b>✓</p> <p>For aluminium, an electron is removed from a <b>3p-sub energy level which has</b></p>                                                                                                                                                                                                                                                                                                              | <p><b>Electron configurations must be written Reject if energy level is written instead of sub energy level</b></p>                                                                   |

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|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
|                      | only one electron hence thermodynamically unstable✓. Lower amount of energy is required to remove the electron. ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Reject the statement neither completely filled nor half filled.                                                                        |
| (d)                  | The <b>beryllium ion</b> in beryllium chloride has a <b>smaller ionic radius</b> ✓ than the <b>magnesium ion</b> in magnesium chloride. The beryllium ion therefore has a higher charge density/ <b>higher polarising power</b> ✓ than the magnesium ion. The chloride ion is <b>more greatly polarised by the beryllium ion than the magnesium ion</b> ✓. This makes <b>beryllium chloride mainly/predominantly covalent</b> ✓ where as sodium chloride is <b>purely ionic</b> . ✓ The ionic bonds in magnesium chloride require a <b>higher amount of energy to be broken</b> ✓ than the covalent bonds in beryllium chloride. | You are required to explain what makes beryllium chloride covalent and magnesium chloride ionic                                        |
| (e)                  | Solubility <b>decreases from beryllium sulphate to barium sulphate</b> . ✓<br><br>This is because <b>cationic radius increases</b> , ✓ <b>both lattice energy and hydration energy of the cations decrease from beryllium ion to barium ion</b> ✓. However, the hydration energy <b>decreases more rapidly</b> ✓ than lattice energy.                                                                                                                                                                                                                                                                                            | No mark is awarded if the statement ...down the group is seen<br>Reject atomic radius increases<br>Emphasize ions for hydration energy |
| (f)                  | Beryllium and aluminium have almost the <b>same atomic radius</b> ✓ and the <b>same electronegativity</b> ✓. Their corresponding <b>ions have almost similar polarising power</b> ✓ and the elements have <b>almost similar standard electrode potentials</b> ✓.                                                                                                                                                                                                                                                                                                                                                                 | This requires to explain reasons that causes diagonal relationship btn Be and Al<br>Emphasize ions for polarizing power.               |
| (g)                  | The <b><math>\pi</math> –electrons in the double bonds in benzene are not localized but delocalised</b> ✓.,<br>This makes benzene very stable and therefore <b>not readily attacked by electron deficient species</b> ✓. ( electrophiles)                                                                                                                                                                                                                                                                                                                                                                                        | Require a good knowledge of the structure of benzene                                                                                   |
| <b>QUESTION FIVE</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                        |
| (a)                  | <b>At constant temperature and pressure, the rate of diffusion of a gas is inversely proportional to the square root of its density.</b> ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Condition of temp and                                                                                                                  |

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|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | pressure shd be specified<br>Accept relative<br>molecular mass<br>instead of<br>density       |
| (b)<br>(i) | $M_r \text{ of } O_2 = (16 \times 2) = 32g$ $\text{Rate of diffusion of } O_2 = \frac{1}{1.87} \text{ min}^{-1}$ $\text{Rate of diffusion of } T = \frac{1}{2.15} \text{ min}^{-1}$ <div style="border-left: 1px solid black; padding-left: 10px; margin-left: 10px;"> <math display="block">\frac{\text{Rate of diffusion of } O_2}{\text{Rate of diffusion of } T} = \frac{\sqrt{RFM \text{ of } T}}{\sqrt{RFM \text{ of } O_2}} \checkmark</math> <math display="block">\left(\frac{1}{1.87}\right) = \frac{\sqrt{RFM \text{ of } T}}{\sqrt{32}} \checkmark</math> <math display="block">\left(\frac{1}{2.15}\right) = \frac{\sqrt{32}}{\sqrt{RFM \text{ of } T}}</math> <math display="block">\left(\frac{2.15}{1.87}\right)^2 = \frac{M_r \text{ of } T}{32}</math> <math display="block">RFM \text{ of } T = 42 \checkmark</math> <math display="block">C_n H_{2n} = 42</math> <math display="block">12n + 2n = 42</math> <math display="block">14n = 42</math> <math display="block">n = 3 \checkmark</math> <p style="text-align: center;"><b>Molecular formula of T is C<sub>3</sub>H<sub>6</sub> ✓</b></p> </div> | Follow the<br>ESCA procedure<br>already<br>explained<br>before                                |
| (ii)       | $CH_3CH = CH_2 \checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Propene ✓                                                                                     |
|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Cyclopropane ✓                                                                                |
| (c)<br>(i) | T is $CH_3CH = CH_2 \checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Accept<br>Propene                                                                             |
| (ii)       | $CH_3CH = CH_2 \xrightarrow{KMnO_4(aq)/\bar{O}H(aq)} CH_3\overset{\overset{OH}{ }}{CH}CH_2OH \quad \checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Reject if heat is<br>included<br>Ignore states<br>Deny if states<br>are included<br>and wrong |



|            |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                            |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| (d)        |                                                                                                                                                                                                                                                                                                                                                                                                                                 | Avoid hanging arrow, arrows shd be curly<br>Mechanism should be continuous                 |
| (e)<br>(i) | Propene is reacted with <b>hydrogen bromide</b> ✓ in presence of an <b>organic peroxide</b> ✓ to form 1-bromopropane                                                                                                                                                                                                                                                                                                            | Reject hydrogen peroxide or just peroxide<br>Accept name of any organic peroxide           |
| (ii)       | 1,2-Dibromoethane is <b>boiled</b> ✓ with an <b>excess alcoholic solution of potassium hydroxide</b> ✓✓ to form <b>ethyne</b> ✓. Ethyne is then reacted with <b>sodium in liquid ammonia</b> ✓ to form <b>sodium acetylide</b> (ethenylsodium)✓ which is then reacted with <b>chloromethane</b> ✓ to form propene.                                                                                                              | You can think of any other alternatives                                                    |
| (iii)      | Propene is reacted with <b>hydrogen bromide</b> ✓ in presence of an <b>organic peroxide</b> ✓ to form <b>1-bromopropane</b> ✓ which is then reacted with <b>sodium in presence dry ether(ethoxyethane)</b> ✓ to form <b>hexane</b> ✓. Hexane is then <b>refluxed with sulphur dichloride dioxide at 40-80°C</b> ✓. using <b>di(benzoyl)peroxide/ di(benzenecarbonyl)peroxide</b> ✓ as catalyst to form <b>1-chlorohexane</b> ✓. | Think of converting propene to any isomer of hex-2-yne then to hexane and proceed as shown |

## QUESTION SIX

11

|            |                                                                                      |                                            |
|------------|--------------------------------------------------------------------------------------|--------------------------------------------|
| (a)<br>(i) | Relative atomic mass of an element is the ratio of the average mass of its naturally | "An atom of carbon-12" shd be clearly seen |
|------------|--------------------------------------------------------------------------------------|--------------------------------------------|

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | occurring atoms to $\frac{1}{12}$ th of the mass of an atom of the Carbon-12 isotope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>in definition</b>                                                                                                                                                                                                      |
| (ii) | Radioisotopes are heavier unstable atoms of the same element with the same number of protons but different number of neutrons that undergo spontaneous disintegration to form lighter stable atoms with emission of radiations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Definition shd bring out radioactivity and isotopy</b>                                                                                                                                                                 |
| b(i) | <p>The <b>spectrometer is evacuated of air</b> ✓ that may cause production of unnecessary ions.</p> <p>The sample to be analysed is introduced into the <b>vapourisation chamber</b> ✓ where it is <b>heated by a tungsten filament to vapourise it</b> ✓.</p> <p>The <b>vapourised sample is then introduced in the ionisation chamber</b> ✓ where it <b>collides with a beam of fast moving electrons</b> ✓ emitted by a hot filament to <b>form positive ions</b> ✓.</p> <p>The positive ions from the ionisation chamber are then <b>accelerated by a strong electric field of varying potentials</b> ✓ in the <b>acceleration chamber</b> ✓ such that only <b>positive ions with the same velocity/kinetic energy</b> ✓ enter the <b>magnetic field</b> ✓ in the <b>deflection chamber</b> ✓.</p> <p>The <b>magnetic field then deflects the ions according to their mass-charge ratio</b> ✓.</p> <p>The <b>strength of the magnetic field is varied such that ions of the same mass-charge ratio are focused onto the detector</b> ✓.</p> <p>The ions are changed into <b>sizeable electric currents which are sent to the amplifier where they are fine tuned</b> ✓ and then to the <b>recorder where they are recorded into peaks that show relative intensities of ions</b> ✓ of a particular mass falling at the detector at that instant.</p> <p>A <b>mass spectrum is thus obtained consisting of a series of peaks of variable intensity</b> ✓ to which mass/charge ratio (m/e) values can be assigned.</p> <p><b>The information from the mass spectrum can then be used to determine RAM, RMM of organic compounds, identify a given compound, analyse mixtures, etc.</b> ✓.</p> | <p><b>Question requires you to name the most essential parts of the modern mass spectrometer and the use of each named part.</b></p> <p><b>Compare approach to question 12</b></p> <p><b>Diagram was not required</b></p> |
| (ii) | <p>Let the percentage abundance of <math>^{20}\text{Q} = y</math> ✓</p> <p>The percentage abundance of <math>^{22}\text{Q} = (100 - y)</math> ✓</p> <p>From, R.A.M = <math>\sum \text{Relative isotopic mass} \times \text{proportion of isotopes}</math> ✓</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>12</p> <p><b>Working should be clearly</b></p>                                                                                                                                                                         |

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|                       | $20.2 = \left(20 \times \frac{y}{100}\right) + \left(22 \times \frac{(100-y)}{100}\right) \checkmark$ $2020 = 20y + 2200 - 22y$ $2y = 180$ $y = 90$ <p>The abundance of <math>^{20}\text{Q}</math> is 90% <math>\checkmark</math> and that of <math>^{22}\text{Q}</math> is 10% <math>\checkmark</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | shown                                                                                         |
| (c)<br>(i)            | <ul style="list-style-type: none"> <li>Uranium-235 is used to release energy in nuclear reactors <math>\checkmark</math></li> <li>Cobalt-60 is used in treatment of cancer <math>\checkmark</math></li> <li>Iodine-131 is used in diagnosis of thyroid diseases</li> <li>Oxygen-18 is used to determine mechanism of esterification of a monocarboxylic acid and a primary alcohol.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Any correct 2<br>Accept any correct other<br>Deduct ½ for each extra wrong                    |
| (ii)                  | It is not effective for atoms that produce similar ions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                               |
| (d)<br>(i)            | <ul style="list-style-type: none"> <li>Atomic number <math>\checkmark</math></li> <li>Mass number <math>\checkmark</math></li> <li>Neutron to proton ratio</li> <li>Binding energy</li> <li>Half-life</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Any correct 2<br>Accept any correct other<br>Deduct ½ for each extra wrong                    |
| (ii)                  | Line <b>A</b> is the stability line $\checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Reject any other                                                                              |
| (iii)                 | Region <b>B</b> is the stability belt/ stability region/ Band of nuclear stability $\checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                               |
| (e)                   | <p>In region 1</p> <p>The isotopes in this region gain stability by <b>reducing the neutron-proton ratio</b>. <math>\checkmark</math><br/>This is achieved by <b>increasing the number of protons and reducing the number of neutrons</b> <math>\checkmark</math> by <b>beta emission</b> <math>\checkmark</math> which <b>reduces the number of neutrons by one and increases the proton number by one</b>. <math>\checkmark</math></p> <p>In region 2.</p> <p>The isotopes in this region gain stability by <b>increasing the neutron-proton</b> <math>\checkmark</math>. This is done by <b>reducing the number of protons and increasing the number of neutrons</b> <math>\checkmark</math> by <b>positron emission or electron capture</b> <math>\checkmark</math> where both processes reduce <b>number of protons by one</b> and <b>increases number of neutrons by one</b>. <math>\checkmark</math></p> | <p>2 Marks</p> <p>2 Marks<br/>Accept either electron capture or positron emission or both</p> |
| <b>QUESTION SEVEN</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                               |
| (a)<br>(i)            | <p><b>Atomic radius increases from Beryllium to barium</b> <math>\checkmark</math> because from Beryllium to Barium, <b>nuclear charge increases, screening effect also increases</b>, <math>\checkmark</math> because an <b>extra energy level completely filled with electrons is added</b> <math>\checkmark</math> from one element</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Spelling is outweigh but not outway                                                           |

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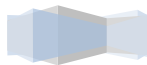


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|            | to the other. <b>Increase in screening effect outweighs increase in nuclear charge. Effective nuclear charge decreases, ✓ and outer most electrons are far and weakly attracted by the nucleus ✓.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                          |
| (ii)       | <b>Electropositivity increases from beryllium to barium ✓</b> because from Beryllium to Barium, <b>nuclear charge increases, screening effect also increases, ✓</b> because an <b>extra energy level completely filled with electrons is added</b> from element to element. <b>Increase in screening effect outweighs increase in nuclear charge. effective nuclear charge decreases, ✓ atomic radius increases ✓ and the valence electrons experience a lower nuclear attraction than repulsion hence easily lost. ✓</b>                                                                                                                                                                                                            |                                                                                                                                                                                                                                          |
| (b)        | Beryllium ion has the <b>smallest</b> ionic radius among the cations ✓<br>Beryllium has the <b>highest</b> electronegativity ✓<br>Beryllium ion has the <b>highest</b> polarising power ✓<br>Beryllium has the <b>least</b> negative standard electrode potential                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Accept any three<br/>Emphasize superlative comparison<br/>Deny ½ mark for each extra wrong answer</b>                                                                                                                                 |
| (c)<br>(i) | <b>Decomposition temperature increases from beryllium carbonate to barium carbonate ✓.</b> This is because as <b>cationic radius increases ✓</b> from the beryllium ion to the barium ion, <b>charge density and polarising power decreases ✓, ionic character increases ✓ and the ions become strongly held ✓</b> leading to an increase in the decomposition temperature.                                                                                                                                                                                                                                                                                                                                                          | <b>Reject if a statement "down the group" is written<br/>Reject ionic radius if the ions are not specified.</b>                                                                                                                          |
| (ii)       | Beryllium carbonate reacts with dilute sulphuric acid to form beryllium sulphate, carbon dioxide and water ✓<br>$\text{BeCO}_3(s) + \text{H}_2\text{SO}_4(aq) \longrightarrow \text{BeSO}_4(aq) + \text{CO}_2(g) + \text{H}_2\text{O}(l) \quad \checkmark \quad \mathbf{1.5}$<br>Magnesium carbonate reacts with dilute sulphuric acid to form magnesium sulphate, carbon dioxide and water. ✓<br>$\text{MgCO}_3(s) + \text{H}_2\text{SO}_4 \longrightarrow \text{MgSO}_4(aq) + \text{CO}_2(g) + \text{H}_2\text{O}(l) \quad \checkmark \quad \mathbf{1.5}$<br>Calcium carbonate, strontium carbonate and barium carbonate react slowly with dilute sulphuric acid and the reaction stops due to formation of insoluble sulphates. ✓ | <b>Products of the reactions shd be given to award<br/>The physical states shd be correct in eqns<br/>Reason why <math>\text{CaCO}_3</math>, <math>\text{SrCO}_3</math> and <math>\text{BaCO}_3</math> don't react shd be specified.</b> |
| (ii)       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Reject sulphuric</b>                                                                                                                                                                                                                  |

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2:00-4:00 pm

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|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p><b>Dilute hydrochloric/nitric acid</b>✓ is separately added to a sample of each of the carbonates in a test tube until <b>no more effervescence/ bubbling/it completely dissolves.</b> ✓</p> <p>Then <b>potassium chromate(VI) solution</b> is added followed by <b>ethanoic acid/sodium hydroxide solution.</b> ✓</p> <p><b>Calcium ions</b> form a <b>yellow precipitate soluble in ethanoic acid/sodium hydroxide solution</b>✓</p> <p><b>Barium ions</b> form a <b>yellow precipitate insoluble in ethanoic acid/ sodium hydroxide solution.</b> ✓</p> <p style="text-align: center;"><b>or</b></p> <p><b>Ammonium oxalate solution</b> is added followed by <b>dilute ethanoic acid</b></p> <p><b>Calcium ions</b> form a <b>white precipitate insoluble in ethanoic acid</b></p> <p><b>Barium ions</b> form a <b>white precipitate soluble in ethanoic acid</b></p> | <p><b>acid</b><br/><b>Reject if carbonates are not dissolved in dilute acid first</b></p> <p><b>All the set of reagents shd be written</b></p> <p><b>Reject calcium/ barium carbonate forms....</b></p> <p><b>Reject use of ammonia solution to distinguish</b></p> |
| (d)                   | <p><b>Magnesium ions</b> react with <b>hydroxyl ions</b>✓ to form <b>insoluble magnesium hydroxide.</b>✓ <b>Magnesium hydroxide is basic</b> hence cannot react with excess sodium hydroxide</p> $\text{Mg}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \longrightarrow \text{Mg}(\text{OH})_2(\text{s})\checkmark$ <p>Ammonia is a <b>weak base which partially ionises to form few hydroxyl ions</b></p> $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^{+}(\text{aq}) + \text{OH}^{-}(\text{aq})\checkmark$ <p>The <b>concentration of the hydroxyl ions formed from the weak ammonia base is low</b>✓ for the <b>ionic product of calcium hydroxide to exceed its solubility product</b>✓. Therefore, <b>no precipitation occurs</b>✓.</p>                                                                                        | <p><b>Reject eqn if it has NH<sub>4</sub>OH</b><br/><b>Eqn must be reversible to award</b></p>                                                                                                                                                                      |
| <b>QUESTION EIGHT</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                     |
| (a)<br>(i)            | <ul style="list-style-type: none"> <li>Osmotic pressure of a dilute solution is directly proportional to the concentration of the solute in a given amount of a solvent provided temperature is constant. ✓.</li> <li>The osmotic pressure of a solution is directly proportional to the absolute temperature provided concentration is constant. ✓.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p><b>Emphasize conditions that are constant</b></p>                                                                                                                                                                                                                |
| (ii)                  | <p>The solution should be dilute✓.</p> <p>The solute particles shouldn't associate or dissociate in the solution✓.</p> <p>Solute should be non-volatile✓.</p> <p>There shouldn't be chemical reaction between the solute and the solvent✓.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p style="text-align: right;">15</p>                                                                                                                                           |

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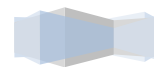
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|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <p>(b)<br/>(i)</p> | <div data-bbox="228 195 1333 569" data-label="Diagram"> </div> <p>A known mass of starch, <math>m(g)</math> is dissolved in a known volume of water✓ to form a solution of known concentration.</p> <p>The starch solution is placed in the outer metal casing which is connected to the piston and the pressure gauge. ✓</p> <p>The starch solution is separated from the water ✓ placed in an inner cylindrical porous tube by a semi-permeable membrane. ✓</p> <p>The piston is pushed down to exert external pressure✓ until the level of the solvent in the capillary tube is just stationary. ✓</p> <p>The pressure at this instant is read off from the pressure gauge and is the osmotic pressure of the solution, <math>\pi</math>. ✓</p> <p>If the temperature of the solution is known, then the relative molecular mass of starch can be determined from the formula,</p> $\pi V = \frac{mRT}{M_r} \checkmark$ <p>Where, <math>\pi</math> is the osmotic pressure.<br/> <math>V</math> is the volume of the solution<br/> <math>m</math> is the mass of starch<br/> <math>R</math> is the molar gas constant<br/> <math>M_r</math> is the relative molecular mass of the polymer.</p> | <p>Well labeled and working diagram<br/>(2 mks)</p>                        |
| <p>(ii)</p>        | <div data-bbox="186 1430 600 1753" data-label="Equation-Block"> <math display="block">m = 2.56g</math> <math display="block">V = 1 \text{ dm}^3 = 1 \times 10^{-3} \text{ m}^3</math> <math display="block">\pi = \left( \frac{0.146}{760} \times 101325 \right) \text{ Nm}^{-2}</math> <math display="block">T = (23 + 273) = 296 \text{ K}</math> <math display="block">R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}</math> <math display="block">\pi V = \frac{m}{M_r} RT \checkmark</math> </div> <div data-bbox="722 1430 1339 1753" data-label="Equation-Block"> <math display="block">M_r = \frac{2.56 \times 8.31 \times 296}{\left( \frac{0.146}{760} \times 101325 \right) \times 1 \times 10^{-3}} \checkmark</math> <math display="block">M_r = 323501.89</math> <math display="block">M_r \approx 323502 \text{ g} \checkmark</math> <p>The molar mass of the polymer is 323502g</p> <math display="block">(C_6H_{10}O_5)_n = 323502</math> <math display="block">[(6 \times 12) + (10 \times 1) + (5 \times 16)]n = 323502</math> </div>                                                                                                                                               | <p>If RFM is used, it shd have no units</p> <p>n shd be a whole number</p> |

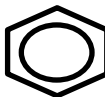
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|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                  |                                                                                                                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|            | $M_r = \frac{mRT}{\pi V}$                                                                                                                                                                                                                                                                                                                                                                                                                                                | $162n = 323502$<br>$n = 1996.9$<br>$n \approx 1997 \checkmark$                                                                                                                                                                                                   |                                                                                                                                |
| (c)<br>(i) | <p>98g of water dissolve 2g of P</p> <p>1000g of water will dissolve <math>\left(\frac{1000 \times 2}{98}\right)</math> g of P</p> <p>= 20.408g of P <math>\checkmark</math></p> <p>Molar mass of P = 323502g</p> <p>323502g of P lower freezing point by 1.86°C</p> <p>20.408g of P lower freezing point by <math>\left(\frac{20.408 \times 1.86}{323502}\right)^\circ\text{C} \checkmark</math></p> <p><math>\Delta T_f = 0.000117^\circ\text{C} \checkmark</math></p> | <p>Depression of freezing point;</p> <p><math>\Delta T_f = T_{f(\text{solvent})}^0 - T_{f(\text{solution})}</math></p> <p><math>T_{f(\text{solution})} = 100 - 0.000117</math></p> <p><math>T_{f(\text{solution})} = 99.9999^\circ\text{C} \checkmark</math></p> |                                                                                                                                |
| (ii)       | <p>There a <b>negligible/ very small change/ almost no change in freezing point of the solution</b> <math>\checkmark</math> since starch has a <b>very high relative molecular mass</b>, <math>\checkmark</math> giving <b>very few dissolved particles in dilute aqueous solutions</b> <math>\checkmark</math> hence a <b>very small freezing point depression</b>. <math>\checkmark</math></p>                                                                         |                                                                                                                                                                                                                                                                  | <p><b>Freezing point depression method is not suitable to determine RFM of polymers</b></p>                                    |
| (d)<br>(i) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                  | <p><b>Scale should be consistent</b></p> <p><b>Axes should be labeled</b></p> <p><b>Straight line graph through origin</b></p> |

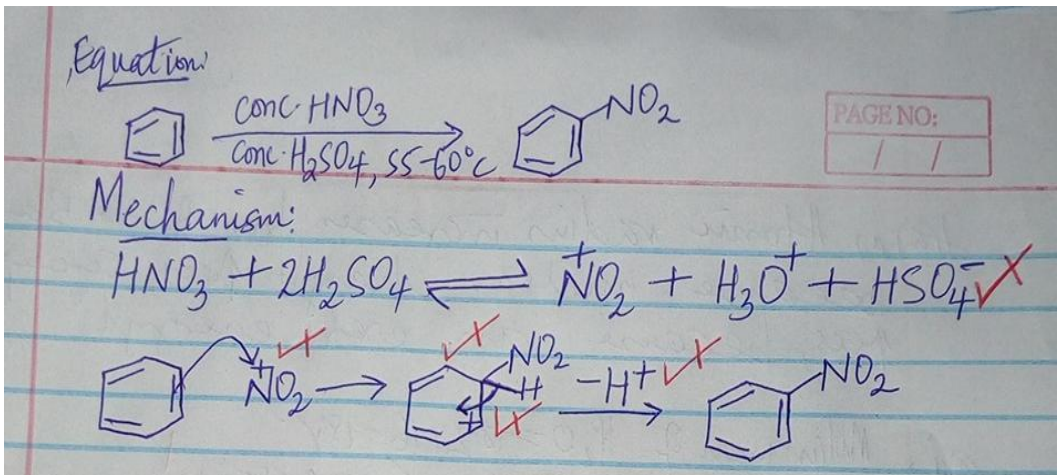
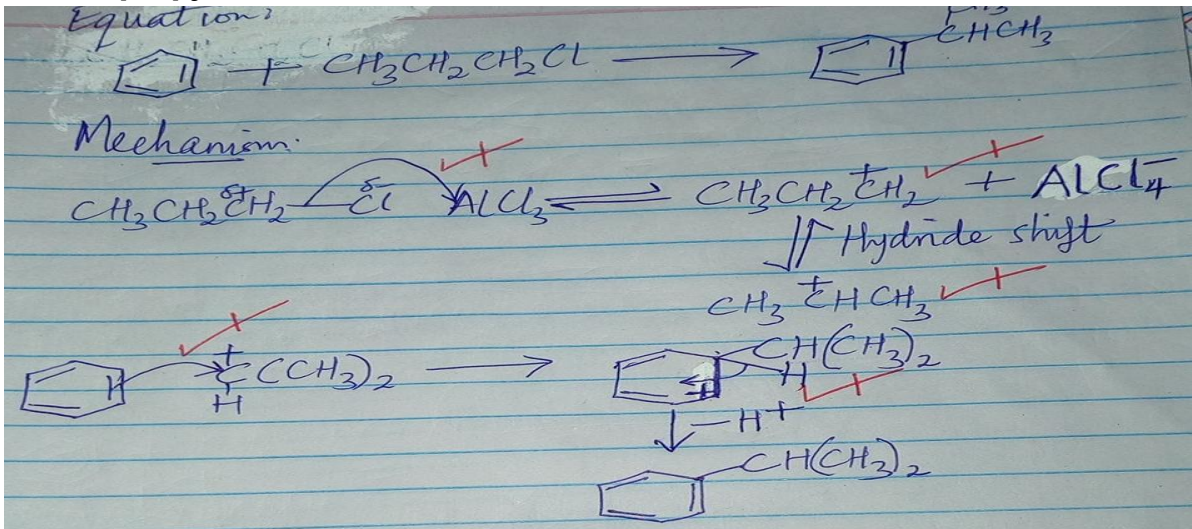
|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                          |                                                        |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------|---|---------|-----|------|-------|------------------|-----------------------------|----------------|-------------------------------------------|------------------------------------------|--|---|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | $\text{Slope} = \frac{620 - 50}{16 - 1.2} = 38.5135 \text{Nm}^{-2} \text{g}^{-1} \text{l} \checkmark$ <p>Since 1 litre = <math>10^{-3} \text{ m}^3</math></p> $38.5135 \text{Nm}^{-2} \text{g}^{-1} \text{l} = 38.5135 \times 10^{-3} \text{Nm}^{-2} \text{g}^{-1} \text{m}^3$ $\text{Slope} = \frac{RT}{RFM} = \frac{8.31 \times 298}{RFM} = 38.5135 \times 10^{-3} \checkmark$ $RFM = \frac{8.31 \times 298}{38.5135 \times 10^{-3}} = 64299 \checkmark$ $\text{Molar mass of } \text{CH}_2 = \text{CHCl} = (2 \times 12) + (3 \times 1) + 35.5 = 62.5 \text{g}$ $62.5n = 64299$ $n = 1028.78 \approx 1029 \checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | n should be a whole number               |                                                        |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| QUESTION NINE  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                          |                                                        |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| (a)<br>(i)     | <p>From 4.05g of <math>\text{H}_2\text{O}</math>;</p> <p>molar mass of <math>\text{H}_2\text{O} = 2 + 16 = 18 \text{g}</math></p> $\text{mass of H} = \frac{2}{18} \times 4.05 = 0.45 \text{g} \checkmark$ <p>From 10080<math>\text{cm}^3</math> of <math>\text{CO}_2</math>;</p> <p>molar mass of <math>\text{CO}_2 = 12 + 32 = 44 \text{g}</math></p> <p>22400 <math>\text{cm}^3</math> of <math>\text{CO}_2</math> at stp contain 44g</p> <p>10080 <math>\text{cm}^3</math> of <math>\text{CO}_2</math> at stp contain <math>\left(\frac{10080 \times 44}{22400}\right) \text{g}</math></p> $= 19.8 \text{g}$ $\text{mass of C} = \frac{12}{44} \times 19.8 = 5.4 \text{g} \checkmark$ <table><tr><td>Elements</td><td>C</td><td>H</td></tr><tr><td>mass(g)</td><td>5.4</td><td>0.45</td></tr></table><br><table><tr><td>moles</td><td><math>\frac{5.4}{12}</math></td><td><math>\frac{0.45}{1} \checkmark</math></td></tr><tr><td>simplest ratio</td><td><math>\frac{\frac{0.45}{12}}{\frac{0.45}{12}}</math></td><td><math>\frac{\frac{0.45}{1}}{\frac{0.45}{12}}</math></td></tr><tr><td></td><td>1</td><td>1 <math>\checkmark</math></td></tr></table> <p>Empirical formula is <math>\text{CH} \checkmark</math></p> | Elements                                 | C                                                      | H | mass(g) | 5.4 | 0.45 | moles | $\frac{5.4}{12}$ | $\frac{0.45}{1} \checkmark$ | simplest ratio | $\frac{\frac{0.45}{12}}{\frac{0.45}{12}}$ | $\frac{\frac{0.45}{1}}{\frac{0.45}{12}}$ |  | 1 | 1 $\checkmark$ | <p>(ii) <math>m = 5.845 \text{g}</math></p> $V = 1.829 \times 10^{-3} \text{m}^3$ $T = (20 + 273) = 293 \text{K}$ $P = 101325 \text{Pa}$ $M_r = \frac{mRT}{PV} \checkmark$ $M_r = \frac{5.845 \times 8.31 \times 298}{101325 \times 1.829 \times 10^{-3}} \checkmark$ $M_r = 78.1$ <p>molar mass of D <math>\approx 78</math></p> $(\text{CH})n = 78$ $(12 + 1)n = 78 \checkmark$ $13n = 78$ $n = 6 \checkmark$ <p>Molecular formula is <math>\text{C}_6\text{H}_6 \checkmark</math></p> |
| Elements       | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | H                                        |                                                        |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| mass(g)        | 5.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.45                                     |                                                        |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| moles          | $\frac{5.4}{12}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{0.45}{1} \checkmark$              |                                                        |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| simplest ratio | $\frac{\frac{0.45}{12}}{\frac{0.45}{12}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\frac{\frac{0.45}{1}}{\frac{0.45}{12}}$ |                                                        |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 $\checkmark$                           |                                                        |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| (b)<br>(i)     | Structural formula<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | IUPAC name<br><br>Benzene                | Try to draw bonds of almost equal length <sup>18</sup> |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| (ii)           | $\text{C}_6\text{H}_6(\text{l}) + \frac{15}{2} \text{O}_2(\text{g}) \longrightarrow 6\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l}) \checkmark \text{ 1.5}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                          | Physical states must be                                |   |         |     |      |       |                  |                             |                |                                           |                                          |  |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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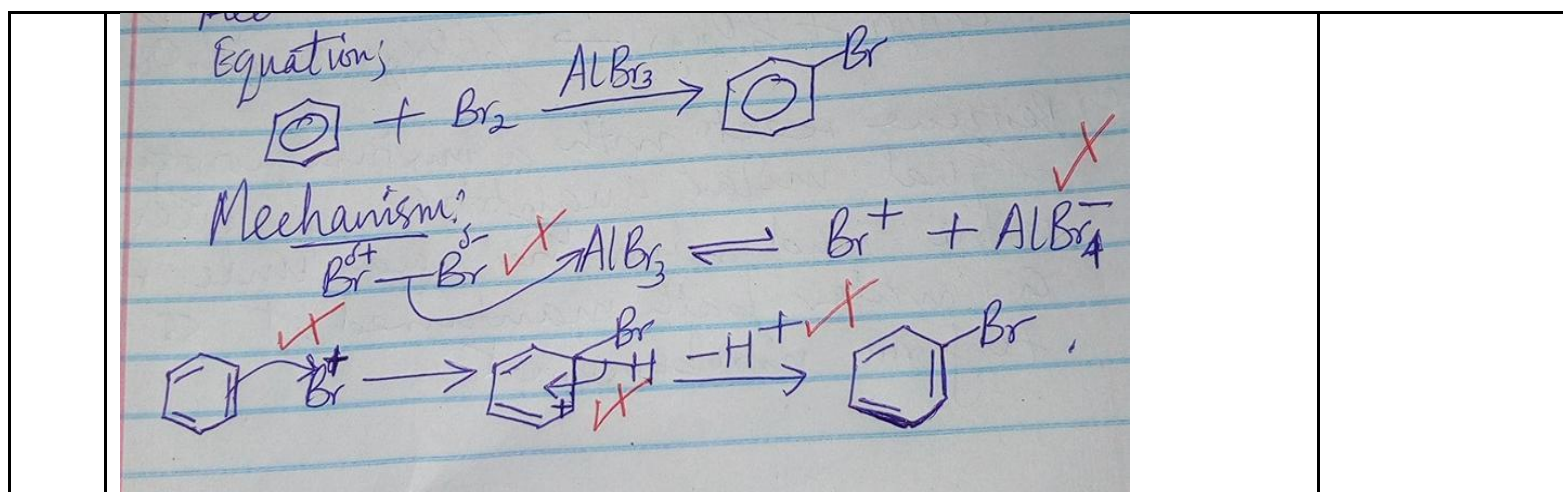
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|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|            |                                                                                                                                                                                                                                                                                                                                 | correct and eqn balanced.                                                                                                           |
| (c)<br>(i) | <p>Benzene reacts with a mixture containing <b>equal molar quantities of concentrated sulphuric acid</b>✓ and <b>concentrated nitric acid</b>✓ under reflux in a water bath maintained at <b>55-60°C</b> ✓ to form <b>nitrobenzene</b>✓.</p>  | <p><b>Reagents, products, conditions and mechanisms shd all be shown</b></p> <p><b>Arrows shdnt be hanging and shd be curly</b></p> |
| (ii)       | <p>Benzene reacts with propylchloride in presence of <b>aluminium chloride catalyst</b>✓ to form <b>propylbenzene</b>✓.</p>                                                                                                                  |                                                                                                                                     |
| (iii)      | <p>Benzene reacts with bromine in presence of <b>aluminium bromide catalyst</b>✓ to form <b>bromobenzene</b>✓.</p>                                                                                                                                                                                                              |                                                                                                                                     |





### QUESTION TEN

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| (a)  | Diagonal relationship is the similarity in properties between elements in the period 2 of the periodic table to their diagonally opposite neighbours in period 3. ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                       |
| (b)  | Beryllium and aluminium have almost the same atomic radius ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Emphasize ions for charge density</b><br><b>Reject any other reason</b>                                            |
| (i)  | Both beryllium and aluminium have the same electronegativity value ✓<br>Both form ions with the same charge density ✓<br>Both beryllium and aluminium have almost the same standard electrode potential. ✓                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                       |
| (ii) | Both elements are rendered passive by concentrated nitric acid. ✓<br>Both react with <b>hot concentrated</b> sodium hydroxide to form complexes. ✓<br>Both form chlorides which are partly covalent and exist as dimers in the <b>vapour phase</b> . ✓<br>Both form carbides which yield methane on hydrolysis. ✓                                                                                                                                                                                                                                                                                                             | <b>Emphasize BOLD words</b><br><br><b>Accept any correct other</b>                                                    |
| (c)  | $\text{Be}(s) + 2\text{OH}^-(aq) + 2\text{H}_2\text{O}(l) \longrightarrow \text{Be}(\text{OH})_4^{2-}(aq) + \text{H}_2(g)$ $\text{or } \text{Be}(s) + 2\text{OH}^-(aq) \longrightarrow \text{BeO}_2^{2-}(aq) + \text{H}_2(g)$ $2\text{Al}(s) + 2\text{OH}^-(aq) + 6\text{H}_2\text{O}(l) \longrightarrow 2\text{Al}(\text{OH})_4^-(aq) + 3\text{H}_2(g)$ $\text{or } 2\text{Al}(s) + 6\text{OH}^-(aq) + 6\text{H}_2\text{O}(l) \longrightarrow 2\text{Al}(\text{OH})_6^{3-}(aq) + 3\text{H}_2(g)$ $\text{or } 2\text{Al}(s) + 2\text{OH}^-(aq) + 2\text{H}_2\text{O}(l) \longrightarrow 2\text{AlO}_2^-(aq) + 3\text{H}_2(g)$ | <b>Accept molecular eqns</b>                                                                                          |
| (d)  | The chloride ion in calcium chloride has a <b>smaller ionic charge</b> ✓ than the sulphate ion in calcium sulphate. This makes the <b>chloride ion less strongly attracted by the calcium ion than the sulphate ion is attracted by the calcium ion</b> ✓ in the ionic lattices of the two compounds. Calcium chloride therefore has a <b>lower lattice dissociation energy than calcium sulphate</b> , ✓ which will require a high amount of                                                                                                                                                                                 | <b>Reject CaCl<sub>2</sub> having a higher hydration energy than CaSO<sub>4</sub></b><br><b>Accept comparison b/n</b> |

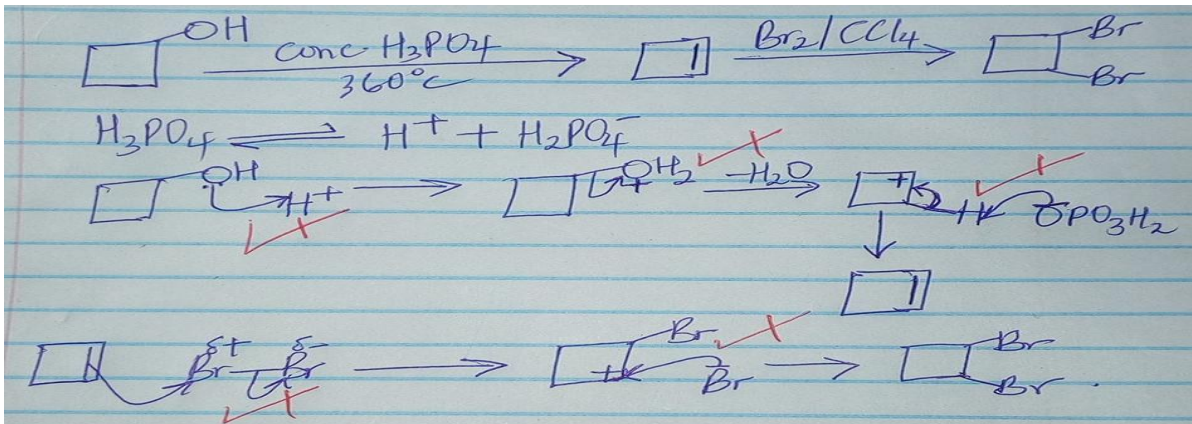
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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|      | energy to break its ionic crystal lattice before hydration occurs.                                                                                                                                        | <b>only lattice enthalpies.</b> |
| (e)  | <b>White solid dissolves</b> ✓ with evolution of <b>bubbles of a colourless gas</b> ✓ to form <b>colourless solution.</b> ✓<br>$Ca_3N_2(s) + 6H_2O(l) \longrightarrow 3Ca(OH)_2(aq) + 2NH_3(g)$ ✓         |                                 |
| (i)  | A <b>yellow precipitate</b> is formed✓<br>$Ba^{2+}(aq) + CrO_4^{2-}(aq) \longrightarrow BaCrO_4(s)$ ✓                                                                                                     |                                 |
| (ii) | <b>White solid dissolves</b> ✓ with evolution of <b>bubbles of a colourless gas</b> ✓ to form <b>colourless solution.</b> ✓<br>$Mg_2C_3(s) + 4HCl(aq) \longrightarrow 2MgCl_2(aq) + CH_3C \equiv CH(g)$ ✓ |                                 |

### QUESTION ELEVEN

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                             |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| (a) | The <b>aluminium ion</b> in aluminium chloride has a <b>higher charge</b> ✓ and a <b>smaller ionic radius</b> ✓ than the <b>sodium ion</b> in sodium chloride. The aluminium ion therefore has a <b>higher charge density</b> and <b>higher polarising power</b> ✓ than the sodium ion. The chloride ion being <b>greatly polarised</b> by the aluminium ion makes <b>aluminium chloride predominantly covalent</b> ✓ hence soluble in <b>non-polar</b> ✓methylbenzene. However, sodium chloride is <b>purely ionic</b> ✓ and <b>insoluble</b> in <b>non-polar</b> ✓ solvents. | <b>Reject if charge, ionic radius, polarizing power or charge density is linked to the compounds instead of the cations</b> |
| (b) | Cyclobutanol is <b>dehydrated</b> ✓ by hot concentrated phosphoric(V) acid to form <b>cyclobutene</b> ✓. <b>Cyclobutene then reacts with bromine</b> ✓ in tetrachloromethane to form <b>1,2-dibromocyclobutane</b> ✓ via an <b>electrophilic addition reaction</b> ✓.<br>                                                                                                                                                                                                                  | <b>Correct names of products for the reactions should be given</b>                                                          |



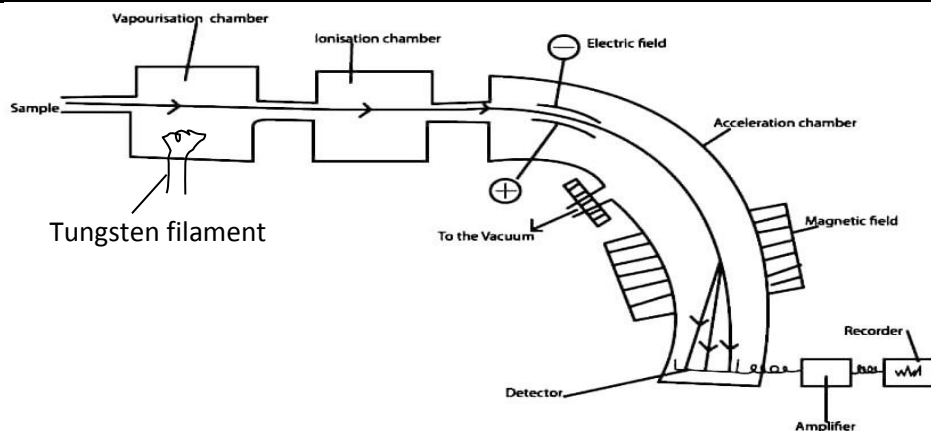
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| (c)                    | <p><b>Actual relative molecular mass of sodium chloride</b> = <math>23 + 35.5 = 58.8</math>✓</p> <p>The obtained relative molecular mass is <b>half the actual value</b>.</p> <p>Sodium chloride being a <b>strong electrolyte</b>, <b>completely ionises</b>✓ in water to form <b>sodium ions and chloride ions</b></p> $NaCl(s) \longrightarrow Na^+(aq) + Cl^-(aq)$ ✓ <p>Therefore <b>1 mole of sodium chloride will form a total of 2 moles of ions in solution</b>✓. Since <b>boiling point elevation is a colligative property</b>, (is directly proportional to the number of non-volatile solute particles in solution) but <b>inversely proportional to relative molecular mass</b>, ✓ the ionisation of sodium chloride <b>increases the boiling point elevation</b>✓ but <b>reduces the relative molecular mass</b>✓ to half the actual value. ✓</p> | <p><b>This question explores the effect of ionization on colligative properties hence RFM</b></p> |
| (d)                    | <p>Iodide ions have a <b>larger ionic radius</b> ✓ and <b>reduce copper(II) ions</b> ✓ to <b>insoluble</b> ✓ <b>copper(I) iodide</b>✓ as they are <b>oxidised to iodine solution</b>✓.</p> $2Cu^{2+}(aq) + 4I^-(aq) \longrightarrow Cu_2I_2(aq) + I_2(aq)$ ✓ <b>1.5</b> <p>Chloride ions have a <b>smaller ionic radius</b> and <b>cannot reduce copper(II) ions</b>✓.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p><b>Accept molecular eqn</b><br/><b>Accept CuI</b></p>                                          |
| (e)                    | <p>Alkenes have <b>carbon-carbon double bonds</b>✓ which are a <b>source of <math>\pi</math> –electrons</b>. ✓ The double bonds are therefore a <b>center of high electron density</b> ✓ and therefore <b>readily attacked by electron deficient species</b>✓ ( electrophiles). Carbonyl compounds have a <b>carbon-oxygen double bond</b> in which the oxygen atom is <b>more electronegative</b> ✓ than the carbonyl carbon atom. The oxygen atom therefore <b>attracts the bonding electrons more towards itself, gaining a partial negative charge</b> ✓ whereas the carbon atom gains a <b>partial positive charge</b>✓. This makes the carbon-oxygen bond polar. The <b>partial positive carbon is readily attacked to the nucleophile</b>, ✓ which adds itself across the carbonyl group to form a single product.</p>                                   | <p><b>You may follow the explanation with an illustration</b></p>                                 |
| <b>QUESTION TWELVE</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                   |
| (a)<br>(i)             | <p>Isotopes are atoms of the same element with the same number of protons but different number of neutrons✓ whereas allotropes are different forms of an element in the same physical state. ✓</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p><b>Accept same atomic number but different mass numbers</b></p>                                |
| (ii)                   | ${}^{12}_6C$ ✓ and ${}^{14}_6C$ ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><b>Accept symbols only. Both atomic no and mass no. shd be shown</b></p>                       |

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Sat 09/10/2021  
2:00-4:00 pm

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(b)  
(i)



Well labeled diagram to earn 2.5 marks.

The procedural flow should be clear from one chamber to another

The **whole apparatus is evacuated of air particles**✓ that may produce unnecessary ions.

The carbon sample is introduced into the **vapourisation chamber**✓ where it is **heated by a tungsten filament to vapourise it**✓.

The **vapourised sample is then introduced in the ionisation chamber**✓ where it is **subjected to a beam of fast moving electrons**✓ emitted by a hot filament. These electrons bombard the atoms of the vapourised sample from which 1 or 2 electrons are removed **to form positive ions**✓.

The positive ions are then **accelerated by a strong electric field of varying potentials**✓ such that only **positive ions with the same velocity/kinetic energy**✓ but with different mass-charge ratio will pass through to the **magnetic field**.✓

The **magnetic field then deflects the ions according to their mass-charge ratio**✓. The **strength of the magnetic field is varied such that ions of the same mass-charge ratio are focused onto the detector**✓.

The ions are changed into **sizeable electric currents which are sent to the amplifier for fine tuning**✓ and then to the **recorder where they are recorded into peaks that show relative intensities of ions**✓ of a particular mass falling at the detector at that instant.

A **mass spectrum is thus obtained consisting of a series of peaks of variable intensity**✓ to which mass/charge ratio (m/e) values can be assigned.

$$R.A.M = \sum \text{Relative isotopic mass} \times \text{proportion of isotope} \checkmark$$

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(ii)

It gives very accurate values of relative atomic mass✓

It uses a small quantity of the sample ✓



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|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------|
| (c) | $\text{Total ratio} = 11 + 9 = 20\checkmark$ $\text{Percentage abundance of A} - 79 = \left(\frac{11}{20} \times 100\right) = 55\checkmark$ $\text{Percentage abundance of A} - 81 = \left(\frac{9}{20} \times 100\right) = 45\checkmark$ $R.A.M = \sum \frac{\text{Relative isotopic mass} \times \text{percentage abundance}}{100}\checkmark$ $R.A.M = \left(\frac{(79 \times 55) + (81 \times 45)}{100}\right)\checkmark$ $R.A.M = 79.9\checkmark$ <p style="text-align: center;"><b>or</b></p> $\text{Total ratio} = 11 + 9 = 20$ $R.A.M = \sum \text{Relative isotopic mass} \times \text{Proportion of isotope}$ $R.A.M = \left(79 \times \frac{11}{20}\right) + \left(81 \times \frac{9}{20}\right)$ $R.A.M = 79.9$ |  |  | <b>Total ratio</b><br><b>Expression</b><br><b>Substitution</b><br><b>Computation</b><br><b>Accuracy/Answer</b> |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------|

|     |                                                                                            |                                                                                                                                                                                                          |                                                                                                                                                                                                                                             |                                                                                                                                             |
|-----|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| (d) | $t_{1/2} = 150 \text{ seconds}$<br>$t = 600 \text{ seconds}$<br>$N_0 = 100\%$<br>$N_t = ?$ | $t_{1/2} = \frac{0.693}{\lambda}$<br>$\lambda = \frac{0.693}{t_{1/2}}$<br>$\lambda = \frac{0.693}{150 \text{ s}}$<br>$\lambda = 0.00462 \text{ s}^{-1}$<br>$\ln\left(\frac{N_0}{N_t}\right) = \lambda t$ | $\ln\left(\frac{100}{N_t}\right) = 0.00462 \times 600$<br>$\ln 100 - \ln(N_t) = 2.772$<br>$\ln(N_t) = 4.60517 - 2.772$<br>$\ln(N_t) = 1.83317$<br>$N_t = 6.25$<br><b>The percentage that decayed is <math>(100 - 6.25) = 93.75\%</math></b> | <b>Accept use of the equation</b><br>$2.303 \log\left(\frac{N_0}{N_t}\right) = \lambda t$<br><br><b>Accept any other logical approaches</b> |
|-----|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|

### QUESTION THIRTEEN

(a)

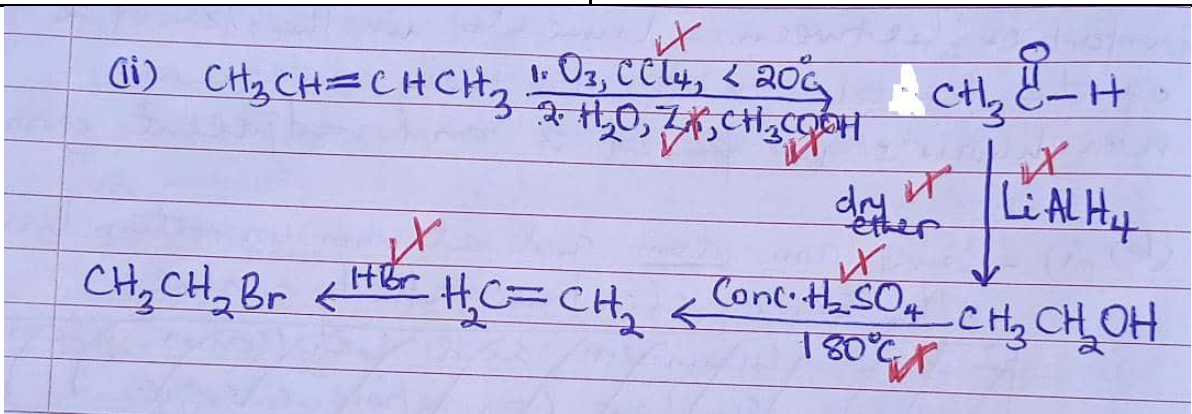
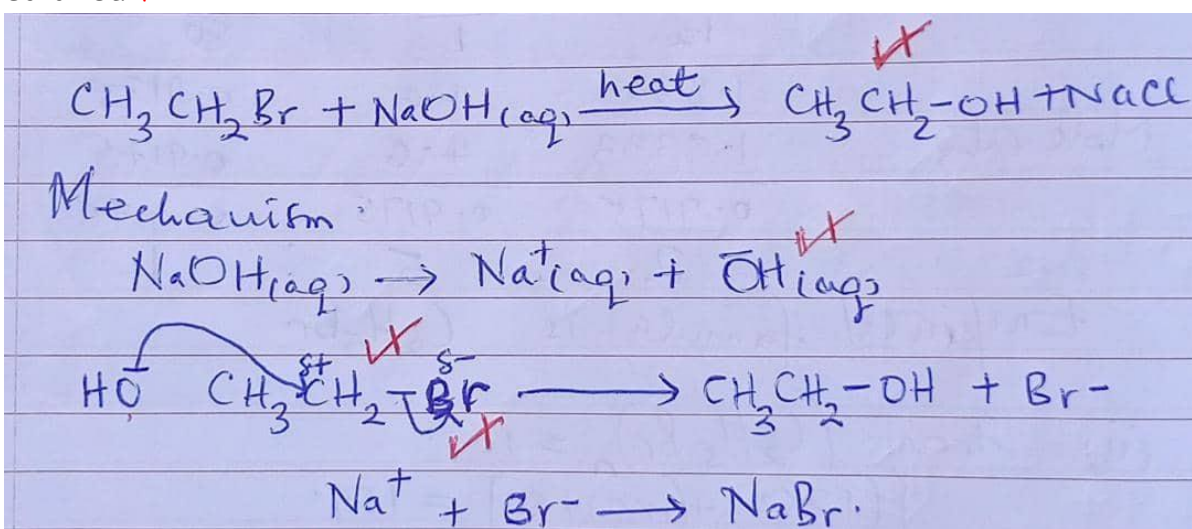
(i)

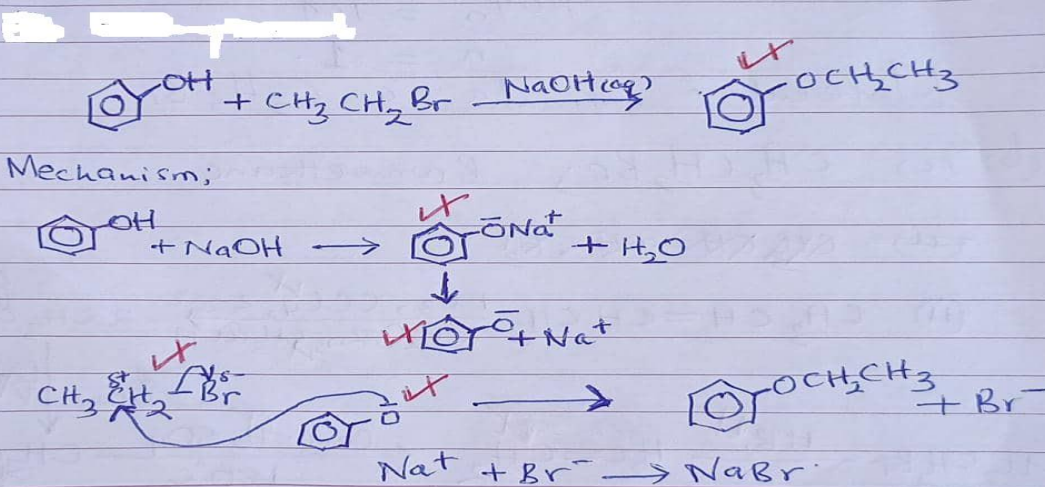
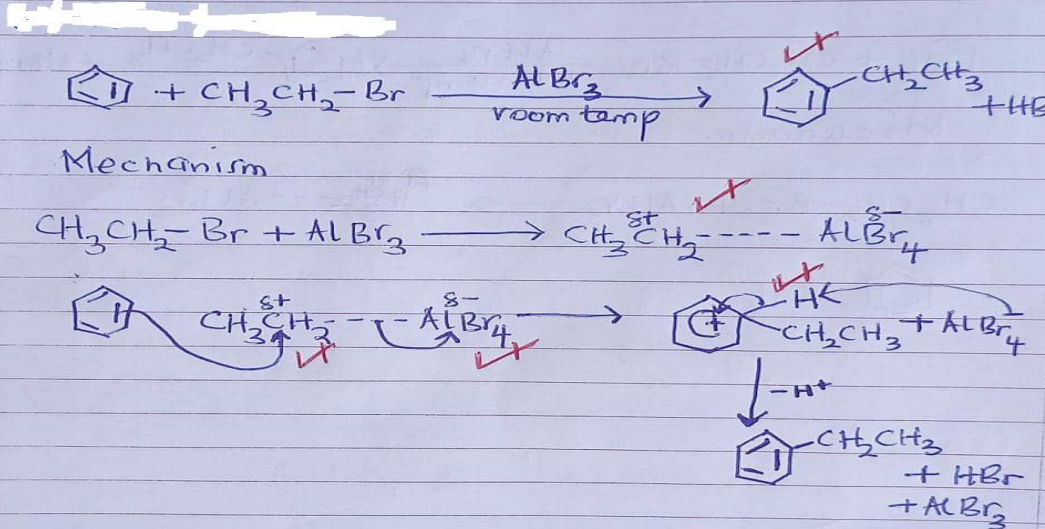
Percentage of bromine =  $100 - (22.0 + 4.6) = 73.4\%$

| Elements | C                       | H               | Br                   |
|----------|-------------------------|-----------------|----------------------|
| Moles    | $\frac{22.0}{12}$       | $\frac{4.6}{1}$ | $\frac{73.4}{80}\%$  |
|          | = 1.8333                | 4.6             | 0.9175               |
| Ratio    | $\frac{1.8333}{0.9175}$ | :               | $\frac{4.6}{0.9175}$ |
|          | 2                       | :               | 5                    |
|          |                         | :               | 1                    |

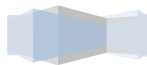
Empirical formula is  $C_2H_5Br$

24

|      |                                                                                                                                                                                                                                                                                                                                                                      |                                       |                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------|
| (ii) | $(C_2H_5Br)n = 109$ $((2 \times 12) + (5 \times 1) + (80 \times 1))n = 109$ $109n = 109$ $n = 1\checkmark$ <p>Molecular formula is <math>C_2H_5Br\checkmark</math></p>                                                                                                                                                                                               |                                       |                                                                |
| (b)  | Structural formula                                                                                                                                                                                                                                                                                                                                                   | IUPAC name ; Bromoethane $\checkmark$ |                                                                |
| (i)  | $CH_3CH_2Br\checkmark$                                                                                                                                                                                                                                                                                                                                               |                                       |                                                                |
| (ii) |                                                                                                                                                                                                                                                                                    |                                       | Chemical symbols must be well written                          |
| (c)  | <p>To the test solutions is added <b>hot aqueous sodium hydroxide solution followed by dilute nitric acid and then silver nitrate solution.</b> <math>\checkmark</math></p> <p>With chloroethane, a <b>white precipitate is observed.</b> <math>\checkmark</math></p> <p>With bromoethane, a <b>pale yellow precipitate is observed.</b> <math>\checkmark</math></p> |                                       | Accept even without dilute nitric acid as part of the reagents |
| (d)  | <p>Bromoethane reacts with <b>hot aqueous sodium hydroxide solution</b><math>\checkmark</math> to form <b>ethanol.</b> <math>\checkmark</math></p>                                                                                                                                                                                                                   |                                       | Conditions must be shown to award.                             |
| (i)  |                                                                                                                                                                                                                                                                                  |                                       |                                                                |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                               |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| (ii)  | <p>Bromoethane reacts with an alkaline solution of phenol to form <b>ethoxybenzene</b>✓.</p>  <p>Mechanism:</p> $\text{C}_6\text{H}_5\text{OH} + \text{NaOH} \rightarrow \text{C}_6\text{H}_5\text{O}^- \text{Na}^+ + \text{H}_2\text{O}$ $\text{C}_6\text{H}_5\text{O}^- \text{Na}^+ + \text{CH}_3\text{CH}_2\text{Br} \rightarrow \text{C}_6\text{H}_5\text{OCH}_2\text{CH}_3 + \text{Br}^-$ $\text{Na}^+ + \text{Br}^- \rightarrow \text{NaBr}$                                                                                                                         | <p><b>Mechanism should proceed by formation of sodium ethoxide first</b></p>  |
| (iii) | <p>Bromoethane reacts with benzene in presence of <b>aluminium bromide catalyst</b> to form <b>ethylbenzene</b>.</p>  <p>Mechanism</p> $\text{CH}_3\text{CH}_2\text{Br} + \text{AlBr}_3 \rightarrow \text{CH}_3\text{CH}_2^+ \text{---} \text{AlBr}_4^-$ $\text{C}_6\text{H}_6 + \text{CH}_3\text{CH}_2^+ \text{---} \text{AlBr}_4^- \rightarrow [\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3]^+ + \text{AlBr}_4^-$ $[\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3]^+ + \text{AlBr}_4^- \rightarrow \text{C}_6\text{H}_5\text{CH}_2\text{CH}_3 + \text{HBr} + \text{AlBr}_3$ | <p><b>Accept any other suitable lewis acid, but its name shd be given</b></p> |

### QUESTION FOURTEEN

|      |                                                                                                                                                                                                                                                            |                                                                                                 |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| (a)  | <p>A primary standard is a substance that is analytically pure and chemically stable such that a known mass of it when weighed is the exact mass that dissolves in water to form a standard solution. ✓</p>                                                |                                                                                                 |
| (i)  |                                                                                                                                                                                                                                                            |                                                                                                 |
| (ii) | <p>A secondary standard is a substance that is not analytically pure and chemically unstable such that a known mass of it when weighed and dissolved in water makes solutions of approximate concentrations hence must be standardized by titration. ✓</p> | <p>26</p>  |

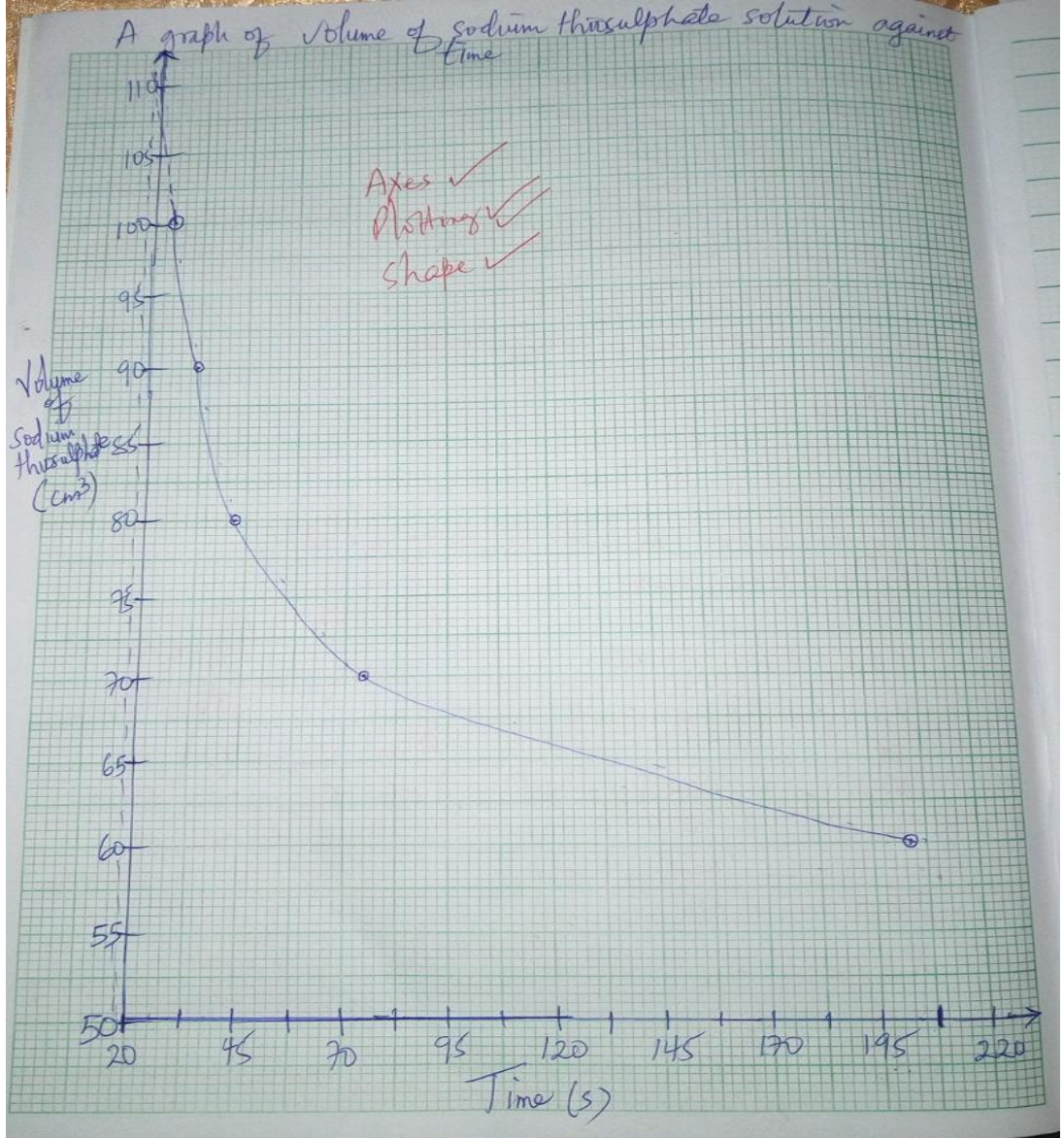
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|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| (b)<br>(i) | <ul style="list-style-type: none"> <li>It is readily soluble in water at room temperature such that all the weighed mass goes into solution. ✓</li> <li>It has a fairly high molar mass such that weighing errors are minimized. ✓</li> <li>It is easily obtainable in a pure state such that the quantity weighed indicates the actual mass present in the standard solution prepared.</li> <li>It is neither deliquescent, efflorescent nor hygroscopic such that the mass weighed is exactly that of the pure sample</li> <li>It is very stable at ordinary temperatures (doesn't decompose easily) hence its chemical nature is not altered.</li> <li>It is able to undergo stoichiometric and instantaneous reactions such that titration errors are minimized.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Accept any correct two |
| (ii)       | <ul style="list-style-type: none"> <li>It is efflorescent ✓</li> <li>When its solution disproportionates when left in air to form sulphur and sulphur dioxide. ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                        |
| (iii)      | Potassium iodate ✓<br>Potassium bromate ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Reject any other       |
| (c)        | <p><b>Moles of <math>\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}</math> in the <math>250\text{cm}^3</math> of the solution</b></p> <p>molar mass of <math>\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} = (23 \times 2) + (11 \times 4) + (16 \times 7) + (10 \times 18) = 382\text{g}</math> ✓</p> <p>moles of <math>\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}</math> in <math>250\text{ cm}^3 = \frac{6.2}{382} = 0.01623</math> ✓</p> <p><b>moles of <math>\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}</math> used during the reaction (moles in <math>25.0\text{cm}^3</math>)</b></p> <p><math>250\text{ cm}^3</math> of solution contain <math>\frac{6.2}{382}</math> moles ✓</p> <p><math>25.0\text{ cm}^3</math> contain <math>\left(\frac{6.2 \times 25.0}{382 \times 250}\right)</math> moles ✓</p> <p><b>moles of the solution that is standardized using equation of the reaction (mole ratio)</b></p> <p><b>From equation;</b></p> $\text{B}_4\text{O}_7^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 5\text{H}_2\text{O}(\text{l}) \longrightarrow 4\text{H}_3\text{BO}_3(\text{aq})$ <p>1 mole of <math>\text{B}_4\text{O}_7^{2-}</math> reacts with 2 moles of <math>\text{H}^+</math> ✓</p> <p><math>\left(\frac{6.2 \times 25.0}{382 \times 250}\right)</math> moles of <math>\text{B}_4\text{O}_7^{2-}</math> react with <math>\left(\frac{6.2 \times 25.0 \times 2}{382 \times 250}\right)</math> moles of acid ✓</p> <p><b>molarity of hydrochloric acid</b></p> <p><math>18.0\text{ cm}^3</math> of acid solution contain <math>\left(\frac{6.2 \times 25.0 \times 2}{382 \times 250}\right)</math> moles ✓</p> <p><math>1000\text{ cm}^3</math> of solution contain <math>\left(\frac{6.2 \times 25.0 \times 2 \times 1000}{382 \times 250 \times 18.0}\right)</math> moles ✓ = <b>0.1803M</b> ✓</p> | Accept a logical flow  |



| (d)<br>(i) | The <b>colourless solution</b> is forms a <b>yellow solid deposit</b> ✓ and <b>bubbles of a colourless gas.</b> ✓<br>$S_2O_3^{2-}(aq) + 2H^+(aq) \longrightarrow S(s) + SO_2(g) + H_2O(l) \checkmark 1.5$                                                                                                                                                                                                                                                                               |                     |                                     |    |     |    |    |    |    |    |    |     |    |  |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------|----|-----|----|----|----|----|----|----|-----|----|--|
| (ii)       | <p>A graph of volume of sodium thiosulphate solution against time</p>  <table><caption>Data points from the graph</caption><thead><tr><th>Time (s)</th><th>Volume of Sodium thiosulphate (cm³)</th></tr></thead><tbody><tr><td>20</td><td>100</td></tr><tr><td>45</td><td>90</td></tr><tr><td>70</td><td>80</td></tr><tr><td>95</td><td>70</td></tr><tr><td>200</td><td>60</td></tr></tbody></table> | Time (s)            | Volume of Sodium thiosulphate (cm³) | 20 | 100 | 45 | 90 | 70 | 80 | 95 | 70 | 200 | 60 |  |
| Time (s)   | Volume of Sodium thiosulphate (cm³)                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |                                     |    |     |    |    |    |    |    |    |     |    |  |
| 20         | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |                                     |    |     |    |    |    |    |    |    |     |    |  |
| 45         | 90                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                                     |    |     |    |    |    |    |    |    |     |    |  |
| 70         | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                                     |    |     |    |    |    |    |    |    |     |    |  |
| 95         | 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                                     |    |     |    |    |    |    |    |    |     |    |  |
| 200        | 60                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                                     |    |     |    |    |    |    |    |    |     |    |  |
| (iii)      | <b>19.5cm³</b> ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>+/-0.5</b><br>28 |                                     |    |     |    |    |    |    |    |    |     |    |  |

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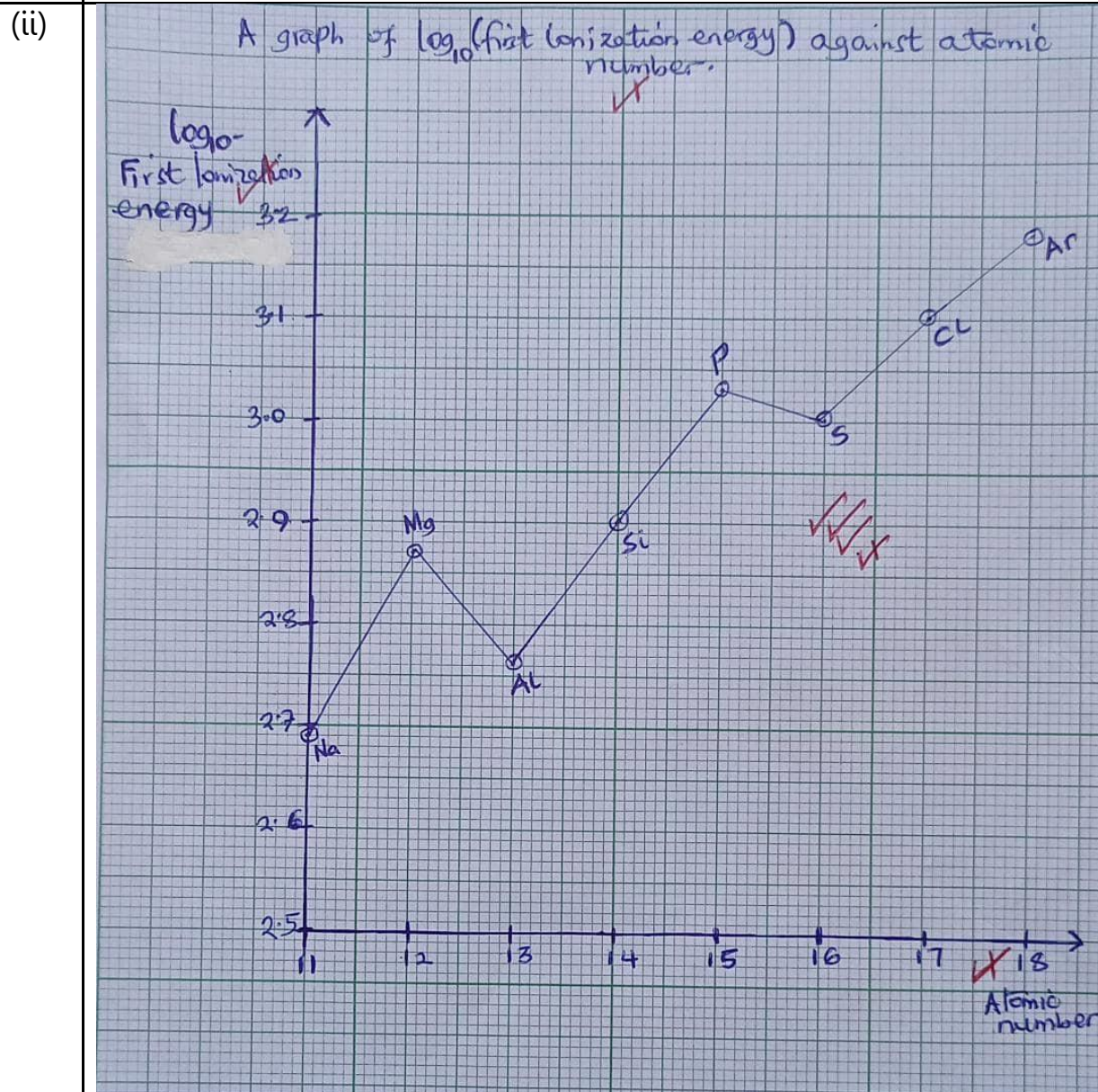
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## QUESTION FIFTEEN

- (a) First ionization energy is the minimum amount of energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of unipositively charged gaseous ions. ✓  
or  
The minimum amount of energy required to remove an electron from a gaseous atom to form a unipositively charged gaseous ion.  
Equation:  $P_{(g)} \longrightarrow P_{(g)}^{+} + e$  ✓

**Using the word one mole makes the electrons, atoms and ions plural.**



**Y-axis shd strictly have values of log ionization energy**

**Axes must be well labeled**

**Deny ½ mark for each incorrectly plotted point**

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                         |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (iii)      | <p>Generally, the <b>first ionization energy increases with increase in atomic number from sodium to argon.</b> ✓</p> <p>This is because <b>nuclear charge increases</b> ✓ because a <b>proton is added to the nucleus from one element to the next</b> and <b>screening effect remains almost constant</b> since <b>electrons are being added to the same energy level</b> ✓. The <b>effective nuclear charge increases</b>, ✓ <b>atomic radius reduces</b> ✓ and the outermost electrons are <b>more strongly attracted to the nucleus</b>, ✓ requiring <b>an increasing amount of energy to be removed</b>.</p> <p style="text-align: center;">Mg: <math>1s^2 2s^2 2p^6 3s^2</math><br/>Al: <math>1s^2 2s^2 2p^6 3s^2 3p^1</math> ✓</p> <p>The <b>first ionization energy of Magnesium is higher than that of Aluminium</b> ✓ because the <b>3p subenergy level of Aluminium has only one electron</b> hence <b>thermodynamically unstable</b> ✓ and a <b>lower amount of energy is required to remove the electron</b> while <b>3s subenergy level of Magnesium is completely filled, thermodynamically stable</b> ✓ and a <b>higher amount of energy is required to remove an electron</b>.</p> <p style="text-align: center;">P: <math>1s^2 2s^2 2p^6 3s^2 3p^3</math><br/>S: <math>1s^2 2s^2 2p^6 3s^2 3p^4</math> ✓</p> <p>Similarly, the <b>first ionization energy of Phosphorous is higher than that of Sulphur</b> ✓ because the <b>3p subenergy level of Phosphorous is half filled with electrons and thus thermodynamically stable</b>. ✓ A <b>higher amount of energy is required to remove an electron</b> from this subenergy level. The <b>3p subenergy level of Sulphur has four electrons and thus thermodynamically unstable</b>. ✓ <b>Lower amount of energy is required to remove an electron from this subenergy level</b>.</p> | <p><b>Both the general trend and any anomalies must be explained</b></p> <p><b>Reject neither completely nor partly filled</b></p> <p style="text-align: center;">"</p> |
| (b)<br>(i) | <p>First ionization energy <b>decreases from lithium to potassium.</b> ✓</p> <p>This is because, from lithium to potassium, <b>nuclear charge increases</b>, ✓ <b>screening effect also increases</b>, ✓ because an <b>extra energy level completely filled with electrons is added from element to element</b>. <b>Increase in screening effect outweighs increase in nuclear charge</b>. <b>Effective nuclear charge decreases</b>, ✓ <b>atomic radius increases</b> ✓ and <b>outermost electron is far and weakly attracted by the nucleus requiring a low amount of energy to be removed</b>. ✓</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p style="text-align: right;">30</p>                                                                                                                                    |
| (ii)       | <p>The reactivity of group of (I) metals increases down the group from Lithium to Potassium as ionisation energy decreases. ✓</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                    |

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|                         | <p>Lithium reacts slowly with cold water to form lithium hydroxide and hydrogen gas✓.</p> $2Li_{(s)} + 2H_2O_{(l)} \longrightarrow 2LiOH_{(aq)} + H_{2(g)}✓$ <p>Sodium reacts with cold water vigorously to form sodium hydroxide and hydrogen gas✓.</p> $2Na_{(s)} + 2H_2O_{(l)} \longrightarrow 2NaOH_{(aq)} + H_{2(g)}✓$ <p>Potassium reacts with cold water explosively (violently) to form potassium hydroxide and hydrogen gas. ✓</p> $2K_{(s)} + 2H_2O_{(l)} \longrightarrow 2KOH_{(aq)} + H_{2(g)}✓$                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                        |
| <b>QUESTION SIXTEEN</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                        |
| (a)<br>(i)              | <p>This is the <b>heat given out</b> when <b>one mole of an ionic solid</b> is <b>formed</b> from its component <b>gaseous ions</b>. ✓</p> <p><b>or</b></p> <p>Lattice energy is the <b>heat absorbed</b> when <b>one mole of an ionic solid</b> is <b>broken</b> down into its component <b>gaseous ions</b>.</p>                                                                                                                                                                                                                                                                                                                                                                                                           | <p><b>Reject if one mole of an ionic solid is missing</b></p> <p><b>Accept ionic crystal lattice</b></p>                                                                                                                                                               |
| (ii)                    | <p>Ionic charge✓</p> <p>Ionic radius✓</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Reject any other</b>                                                                                                                                                                                                                                                |
| (b)                     | <p>Ionic charge;</p> <p>The <b>higher product of ionic charges</b>, ✓ the <b>higher the lattice energy</b> because of <b>strong electrostatic forces of attraction</b>✓ between the oppositely charged ions.</p> <p>Ionic radius</p> <p>The <b>smaller the ionic radii</b>✓ of the oppositely charged ions, the <b>higher the lattice energy</b> because the ions will <b>attract each other strongly</b>✓.</p>                                                                                                                                                                                                                                                                                                              | <b>Accept vice versa explanations</b>                                                                                                                                                                                                                                  |
| (c)<br>(i)              | <p>Diagram illustrating the Born-Haber cycle for the formation of RbCl(s) from its elements:</p> $Rb(s) + \frac{1}{2}Cl_2(g) \xrightarrow{\Delta H_{f,RbCl}} RbCl(s)$ <p>The cycle involves the following steps and enthalpy changes:</p> <ul style="list-style-type: none"> <li><math>Rb(s) \rightarrow Rb(g)</math>: <math>\Delta H_{atm,Rb}</math></li> <li><math>Rb(g) \rightarrow Rb^+(g)</math>: <math>\Delta H_{1st\ IE}</math></li> <li><math>\frac{1}{2}Cl_2(g) \rightarrow Cl(g)</math>: <math>\Delta H_{atm,Cl_2}</math></li> <li><math>Cl(g) \rightarrow Cl^-(g)</math>: <math>\Delta H_{1st\ EA}</math></li> <li><math>Rb^+(g) + Cl^-(g) \rightarrow RbCl(s)</math>: <math>\Delta H_{lattice}</math></li> </ul> | <p><b>The cycle should have well balanced equations</b></p> <p><b>Any ions present should be gaseous and have correct charges</b></p> <p><b>The chemical symbols of the elements must be correct</b></p> <p><b>State symbols must be shown and must be correct</b></p> |

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| (ii)       | $\Delta H_{f,RbCl}^{\theta} = \Delta H_{atm,Rb}^{\theta} + \Delta H_{1st IE}^{\theta} + \Delta H_{atm,Cl_2}^{\theta} + \Delta H_{1stEA} + \Delta H_{lattice}^{\theta} \checkmark$ $-431 = +84 + 397 + \frac{242}{2} + \Delta H_{1stEA} + (-675) \checkmark$ $\Delta H_{1stEA} = -431 - 84 - 397 - 121 + 675$ $\Delta H_{1stEA} = -358 \text{ kJmol}^{-1} \checkmark$ | <b>ESCA</b><br><br><b>Reject -358, -358 or if no units</b>                                                      |
| (d)<br>(i) | $\Delta H_{solution} = \Delta H_{lattice} + \Delta H_{hydration} \checkmark$ $\Delta H_{solution} = +675 + \left( \Delta H_{hydration} + \Delta H_{hydration} \right) \checkmark$ $\Delta H_{solution} = +675 + (-301 + -364) \checkmark$ $\Delta H_{solution} = +10 \text{ kJmol}^{-1} \checkmark$                                                                  | <b>Lattice energy used in the substitution here should be positive</b><br><b>Reject +10, +10 or if no units</b> |
| (ii)       | <p>Rubidium chloride has a <b>low solubility in water</b> since its <b>enthalpy of solution is positive(endothermic)</b> <math>\checkmark</math></p>                                                                                                                                                                                                                 | <b>Reject insoluble, sparingly soluble, lower solubility</b>                                                    |

